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chapter

2

Stoichiometry and Bacterial Energetics

Probably the most important concept in the engineering design of systems for biological treatment is the *mass balance.* For a given quantity of waste, a mass balance is used to determine the amount of chemicals that must be supplied to satisfy the energy, nutrient, and environmental needs of the microorganisms. In addition, the amounts of end products generated can be estimated. Examples of chemicals are oxygen as an electron acceptor, nitrogen and phosphorus as nutrients for biomass growth, and lime or sulfuric acid to maintain the pH in the desired range. Examples of end products of importance are excess microorganisms (sludge), which is a costly disposal problem, and methane from anaerobic systems, which can be a useful source of energy.

Balanced chemical equations are based upon the concept of *stoichiometry,* which is an aspect of chemistry concerned with mole relationships among reactants and products in chemical reactions. Although anyone who has taken even the most rudimentary chemistry course is familiar with writing balanced reactions, several features inherent to microbial reactions complicate the stoichiometry. First, microbial reactions often involve oxidation and reduction of more than one species. Second, the microorganisms have two roles, as catalysts for the reaction, but also as products of the reaction. Third, microorganisms carry out most chemical reactions in order to capture some of the energy released for cell synthesis and for maintaining cellular activity. For this reason, we must consider reaction energetics, as well as balancing for elements, electrons, and charge.

We take an approach that is fundamentally based and also very useful for practice. In this chapter, we demonstrate how to write stoichiometric equations that balance elements, electrons, charge, and energy. This approach integrates all the factors that control microbial growth and its relationships to the materials that the cells consume and produce.

2.1 An Example Stoichiometric Equation

One of the first balanced equations for biological oxidation of a wastewater was that presented by Porges, Jasewicz, and Hoover (1956) for a casein-containing wastewater:

$$
\begin{array}{lcccccc}
 & \underset{\text{“casein”}}{C_8H_{12}O_3N_2} + & 3\ O_2 \rightarrow & \underset{\text{“bacterial cells”}}{C_5H_7O_2N} + & NH_3 + & 3\ CO_2 + & H_2O \\
\text{Formula Weight:} & 184 & 96 & 113 & 17 & 132 & 18 \\
 & \sum = 280 & & \sum = 280 & & &
\end{array}
\qquad [2.1]
$$

Equation 2.1 indicates that, for each 184 g of casein consumed by microorganisms, 96 g of oxygen must be supplied for the reaction to proceed properly. The reaction produces 113 g of new microbial cells, 17 g of ammonia (or 14 g of ammonia-N), 132 g of carbon dioxide, and 18 g of water. Such knowledge is essential for designing a biological treatment system for casein treatment. For example, when we treat 1,000 kg/d of casein, we must supply 520 kg/d of oxygen via aeration, and 610 kg/d of biomass solids (i.e., dry sludge) must be dewatered and disposed of.

In Equation 2.1, some of the carbon in casein is fully oxidized to CO_2. Thus, casein is the electron-donor substrate. The rest of the carbon in casein is incorporated into the newly synthesized biomass, because casein also is the carbon source. The complex protein-containing mixture in casein is represented through the empirical formula $C_8H_{12}O_3N_2$. This formula was constructed from knowledge of the relative mass proportions of organic carbon, hydrogen, oxygen, and nitrogen contained in the wastewater, values that can be obtained from normal organic chemical analyses for each element present.

The same approach is taken for the empirical formula for bacterial cells of $C_5H_7O_2N$. Bacterial cells are highly complex structures containing a variety of carbohydrates, proteins, fats, and nucleic acids, some with very high molecular weights. Indeed, microorganisms contain many more than the four elements indicated by the above equation, such as phosphorus, sulfur, iron, and many other elements that are generally present in only trace amounts. An empirical formula could contain as many elements as desired, as long as the relative proportions on a mass basis are known. However, Porges et al. (1956) chose to represent only the four major elements, and this is generally satisfactory for most practical purposes. The mass requirements for elements not shown in the formula can be determined once the mass of bacterial cells formed from a given reaction is known. For example, phosphorus normally represents about 2 percent of the bacterial organic dry weight. Thus, if 1,000 kg/d of casein were consumed in a treatment process, then 0.02(610) or 12 kg/d of phosphate-phosphorus must be present in the wastewater or added to it in order to satisfy bacterial needs for this element.

Porges et al. (1956) obtained Equation 2.1 from empirical data. Could we predict the stoichiometry of such a reaction? The answer is yes, and the rest of the chapter develops and applies an approach for doing this. To achieve this goal, we need three things:

1. Empirical formula for cells
2. Framework for describing how the electron-donor substrate is partitioned between energy generation and synthesis

3. Means to relate the proportion of the electron-donor substrate that is used to synthesize new biomass to the energy gained from catabolism and the energy needed for anabolism.

These three items are presented in order in the rest of this chapter.

2.2 EMPIRICAL FORMULA FOR MICROBIAL CELLS

The empirical formula noted earlier for cells ($C_5H_7O_2N$) was one of the first used in the balancing of biological reactions. However, the relative proportion of elements actually present in cells depends on the characteristics of the microorganisms involved, the substrates being used for energy, and the availability of other nutrients required for microbial growth. If microorganisms are grown in a nitrogen-deficient environment, they tend to produce more fatty material or carbohydrates, and the resulting empirical cell formula should reflect a smaller proportion of nitrogen. Table 2.1 is a summary of empirical cell formulas reported by others, some for anaerobic and some for aerobic growth, some for mixed cultures of microorganisms and others for pure cultures, and some for cultures grown on different organic substrates. The nitrogen content ranges from 6 to 15 percent, although it averages the "typical" 12 percent.

An extremely important way to compare empirical cell formulas is by the ratio of oxygen required for full oxidation of the cellular carbon per unit weight of cells. This oxygen requirement is termed *COD′*, which we define as the *calculated oxygen demand.* This normally is equal to the *chemical oxygen demand (COD),* which is a standard chemical procedure for evaluating this quantity and is based on the reduction of dichromate in a boiling acid solution. The COD′ can be found from the empirical formula by

$$C_nH_aO_bN_c + \left(\frac{2n + 0.5a - 1.5c - b}{2}\right) O_2 \rightarrow nCO_2 + cNH_3 + \frac{a - 3c}{2} H_2O \quad \textbf{[2.2]}$$

and

$$\text{COD}'/\text{Weight} = \frac{(2n + 0.5a - 1.5c - b)16}{12n + a + 16b + 14c} \quad \textbf{[2.3]}$$

where

$$n = \%\text{C}/12\text{T},\ a = \%\text{H}/\text{T},\ b = \%\text{O}/16\text{T},\ \text{and}\ c = \%\text{N}/14\text{T}$$

and

$$\text{T} = \%\text{C}/12 + \%\text{H} + \%\text{O}/16 + \%\text{N}/14 \quad \textbf{[2.4]}$$

If the mass distribution of the four different organic elements in a biological culture is known, then the empirical formula for the cells can readily be constructed and the COD′ computed.

Table 2.1 Empirical chemical formulas for prokaryotic cells

Empirical Formula	Formula Weight	COD′ Weight	% N	Reference	Growth Substrate and Environmental Conditions
Mixed Cultures					
$C_5H_7O_2N$	113	1.42	12	1	casein, aerobic
$C_7H_{12}O_4N$	174	1.33	8	2	acetate, ammonia N source, aerobic
$C_9H_{15}O_5N$	217	1.40	6	2	acetate, nitrate N source, aerobic
$C_9H_{16}O_5N$	218	1.43	6	2	acetate, nitrite N source, aerobic
$C_{4.9}H_{9.4}O_{2.9}N$	129	1.26	11	3	acetate, methanogenic
$C_{4.7}H_{7.7}O_{2.1}N$	112	1.38	13	3	octanoate, methanogenic
$C_{4.9}H_9O_3N$	130	1.21	11	3	glycine, methanogenic
$C_5H_{8.8}O_{3.2}N$	134	1.16	10	3	leucine, methanogenic
$C_{4.1}H_{6.8}O_{2.2}N$	105	1.20	13	3	nutrient broth, methanogenic
$C_{5.1}H_{8.5}O_{2.5}N$	124	1.35	11	3	glucose, methanogenic
$C_{5.3}H_{9.1}O_{2.5}N$	127	1.41	11	3	starch, methanogenic
Pure Cultures					
$C_5H_8O_2N$	114	1.47	12	4	bacteria, acetate, aerobic
$C_5H_{8.33}O_{0.81}N$	95	1.99	15	4	bacteria, undefined
$C_4H_8O_2N$	102	1.33	14	4	bacteria, undefined
$C_{4.17}H_{7.42}O_{1.38}N$	94	1.57	15	4	*Aerobacter aerogenes,* undefined
$C_{4.54}H_{7.91}O_{1.95}N$	108	1.43	13	4	*Klebsiella aerogenes*, glycerol, $\mu = 0.1\ h^{-1}$
$C_{4.17}H_{7.21}O_{1.79}N$	100	1.39	14	4	*Klebsiella aerogenes,* glycerol, $\mu = 0.85\ h^{-1}$
$C_{4.16}H_8O_{1.25}N$	92	1.67	14	5	*Escherichia coli,* undefined
$C_{3.85}H_{6.69}O_{1.78}N$	95	1.30	15	5	*Escherichia coli,* glucose
Highest	218	1.99	15		
Lowest	92	1.16	6		
Median	113	1.39	12		

References: [1]Porges et al. (1956); [2]Symons and McKinney (1958); [3]Speece and McCarty (1964); [4]Bailey and Ollis (1986); [5]Battley (1987).

Example 2.1

EMPIRICAL BIOMASS FORMULA A sample of a biological culture is submitted to a chemical laboratory for analysis of the percentage by weight of each major element present in the organic portion. The laboratory evaporates the sample to dryness and then places it in an oven overnight at 150 °C to drive off all water present. The organic portion of the residue remaining is then analyzed, after which the sample is burned in a Muffle furnace at 550 °C to determine the weight of the ash remaining. The ash contains the phosphorus, sulfur, iron, and other inorganic elements present in the sample. The composition of the cells by weight is found to be 48.9% C, 5.2% H, 24.8% O, 9.46% N, and 9.2% ash. Prepare an empirical formula for the cells, letting $c = 1$, and determine the COD′/organic weight ratio for the cells.

$$T = 48.9/12 + 5.2 + 24.8/16 + 9.46/14 = 11.50$$

and

$$n = 48.9/(12 \times 11.5) = 0.354, a = 5.2/11.5 = 0.452,$$
$$b = 24.8/(16 \times 11.5) = 0.135, \text{ and } c = 9.46/(14 \times 11.5) = 0.0588.$$

In order to normalize to $c = 1$, divide by 0.0588, in which case the following result is obtained as the empirical formula for the cells:

$$C_{6.0}H_{7.7}O_{2.3}N$$

This formula illustrates that fractional coefficients (e.g., the 2.3 on O) are normal and correct. Rounding fractional coefficients to whole numbers introduces errors and should not be done. The COD′ to organic weight ratio is then equal to

$$(2 \times 6.0 + 0.5 \times 7.7 - 1.5 - 2.3)16/(12 \times 6 + 7.7 + 16 \times 2.3 + 14) = 1.48 \text{ g COD}'/\text{g cells}$$

2.3 SUBSTRATE PARTITIONING AND CELLULAR YIELD

When microorganisms use an electron-donor substrate for synthesis, a portion of its electrons (f_e^0) is initially transferred to the electron acceptor to provide energy for conversion of the other portion of electrons (f_s^0) into microbial cells, as illustrated in Figure 2.1. The sum of f_s^0 and f_e^0 is 1. Cells also decay because of normal maintenance or predation; part of the electrons in f_s^0 are transferred to the acceptor to generate more energy, and another part is converted into a nonactive organic cell residue.

The portions initially converted into cells, f_s^0, and used to generate energy, f_e^0, provide the framework for partitioning the substrate between energy generation and synthesis. A very important facet of the partitioning framework is that it is in terms

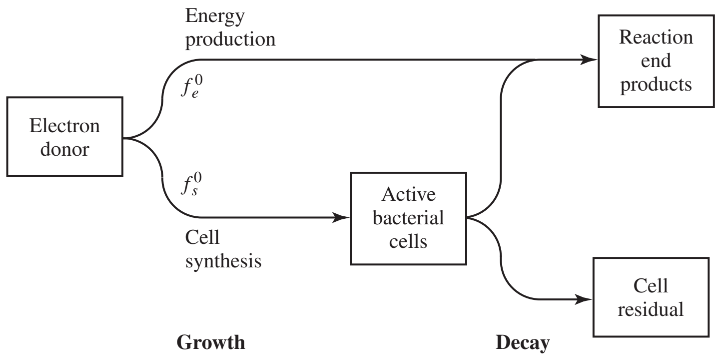

Figure 2.1 Utilization of electron donor for energy production and synthesis.

of electron equivalents (e^- eq). Because electron flows generate the cell's energy, it is essential that the partitioning be expressed as electron equivalents.

The fraction f_s^0 can be converted into mass units, such as g cell produced/g COD′ consumed. When expressed in mass units, it is termed the true yield and given the symbol Y. The conversion from f_s^0 to Y is

$$Y = f_s^0(M_c \text{ g cells/mol cells})/[(n_e e^- \text{ eq/mol cells})(8 \text{ g COD}/e^- \text{ eq donor})]$$

in which $M_c =$ is the empirical formula weight of cells, n_e is the number of electron equivalents in an empirical mole of cells, and the donor mass is expressed as COD. When cells are represented by $C_5H_7O_2N$ and ammonium is the nitrogen source, $M_c = 113$ g cells/mol cells, and $n_e = 20$ e^- eq/mol cells. Then, the conversion is $Y = 0.706 f_s^0$, and Y is in g cells/g COD. The numbers used in the conversion change if the cell formula differs or if the cells use oxidized nitrogen sources, such as NO_3^-; these changes are discussed later.

Mathematically, the growth rate of microbial cells is frequently expressed as

$$\frac{dX_a}{dt} = Y\left(\frac{-dS}{dt}\right) - bX_a \qquad \textbf{[2.5]}$$

where dX_a/dt represents the net growth rate (M/L^3T) of active organism (X_a, M/L^3), $-dS/dt$ represents the rate of disappearance (M/L^3T) of substrate (S, M/L^3), b is the decay rate (T^{-1}) of the organisms, and Y is the true yield of microorganisms (M/M). (A full development of Equation 2.5 and the equations that follow is given in Chapter 3. A short development is used here to support the concept of yield.) The net growth rate is equal the difference between the growth from substrate consumption minus decay due to self (endogenous) respiration or predation. The net yield (Y_n, M/M) can be found by dividing Equation 2.5 by the rate of substrate utilization:

$$Y_n = \frac{dX_a/dt}{-dS/dt} = Y - b\frac{X_a}{-dS/dt} \qquad \textbf{[2.6]}$$

The net yield is less than Y, because some of the electrons originally present in the substrate must be consumed for energy of maintenance. When considering net yield, the portion of electrons used for synthesis is f_s rather than f_s^0, and the portion for energy generation is f_e rather than f_e^0. Still, the sum of f_s and f_e equals 1, and $f_s < f_s^0$, while $f_e > f_e^0$.

Equation 2.6 indicates that this decayed portion becomes larger as the decay rate or organism concentration increases or as the rate of substrate consumption declines. For the casein example (Equation 2.1), the cellular yield of 0.61 g cells/g casein actually represents a net yield, as it was developed from experimental data in which a mixed culture of organisms was growing from substrate utilization and, at the same time, decaying.

When the rate of substrate utilization per unit mass of cells is sufficiently low, the right side of Equation 2.6 can become zero, meaning that the net yield of cells, Y_n, also equals zero. The substrate utilization rate is then just sufficient to maintain the cells, and no net growth of active cells results. Under these conditions:

$$Y_n = 0, \quad \frac{-dS/dt}{X_a} = \frac{b}{Y} = m \qquad [2.7]$$

For this case, the substrate utilization rate per unit mass of organisms is termed the maintenance energy (m, M/MT). From Equation 2.7, m is proportional to b and inversely proportional to Y. When the substrate utilization rate is less than m, the substrate available is insufficient to satisfy the total metabolic needs of the microorganisms. This represents a form of starvation.

Example 2.2

GROWTH RATE AND NET YIELD A reactor containing 500 mg/L of active microorganisms is consuming acetate at a rate of 750 mg L^{-1} d^{-1}. For this culture, $Y = 0.6$ g cells per g acetate and $b = 0.15$ d^{-1}. Determine the specific growth rate of the microorganisms $[(dX_a/dt)/X_a]$, the specific rate of substrate utilization $[(-dS/dt)/X_a]$, and the net yield of cells.

From Equation 2.5, $(dX_a/dt)/X_a = [Y(-dS/dt)/X_a - b] = [0.6(750)/500] - 0.15 = 0.75$ d^{-1}. This means that the population of organisms under these conditions is increasing at a rate of 75 percent per day. Next, $(-dS/dt)/X_a = 750/500 = 1.5$ g acetate per g cells per day. In other words, the microorganisms are consuming 1.5 times their weight in food each day. The net yield is given by Equation 2.6, $Y_n = 0.6 - 0.15/1.5 = 0.5$ g cells per g acetate per day. This is 0.5/0.6 or 83 percent of the true yield Y.

2.4 ENERGY REACTIONS

Microorganisms obtain their energy for growth and maintenance from oxidation-reduction reactions. Even the photosynthetic microorganisms, which obtain energy from electromagnetic radiation or sunlight, use oxidation-reduction reactions to convert the light energy into ATP and NADH.

Oxidation-reduction reactions always involve an electron donor and an electron acceptor. Generally, we think of the electron donor as being the "food" substrate for the organisms. The most common electron donor for all nonphotosynthetic organisms, except some prokaryotes, is organic matter. The chemolithotrophic prokaryotes, however, use reduced inorganic compounds, such as ammonia and sulfide, as electron donors in energy metabolism. Prokaryotes are thus exceptionally versatile. The common electron acceptor under aerobic conditions is diatomic or molecular oxygen (O_2). However, under anaerobic conditions, some prokaryotes can use other electron acceptors in energy metabolism, including nitrate, sulfate, and carbon dioxide. In some cases, organic matter is used as the electron acceptor, as well as the electron donor, and the reaction is then termed fermentation.

Examples of different energy reactions in which glucose is the electron donor, listed below, show that the energy gained from one mole of glucose varies widely, depending on the electron acceptor.

	Free Energy kJ/mol glucose	
Aerobic oxidation: $C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O$	−2,880	**[2.8]**
Denitrification: $5\,C_6H_{12}O_6 + 24\,NO_3^- + 24\,H^+ \rightarrow 30\,CO_2 + 42\,H_2O + 12\,N_2$	−2,720	**[2.9]**
Sulfate reduction: $2\,C_6H_{12}O_6 + 6\,SO_4^{2-} + 9\,H^+ \rightarrow 12\,CO_2 + 12\,H_2O + 3\,H_2S + 3\,HS^-$	−492	**[2.10]**
Methanogenesis: $C_6H_{12}O_6 \rightarrow 3\,CO_2 + 3\,CH_4$	−428	**[2.11]**
Ethanol Fermentation: $C_6H_{12}O_6 \rightarrow 2\,CO_2 + 2\,CH_3CH_2OH$	−244	**[2.12]**

Obviously, microorganisms would like to obtain as much energy from a reaction as possible; therefore, they would prefer to use oxygen as an electron acceptor. Not all microorganisms can use oxygen as an electron acceptor, and these anaerobic microorganisms are unable to compete with the aerobes when oxygen is available. On the other hand, when oxygen is absent, anaerobic microorganisms may dominate. The order of preference for electron acceptors based upon energy considerations alone would be oxygen, nitrate, sulfate, carbon dioxide (methanogenesis), and, finally, fermentation. Some organisms, such as *E. coli,* have the ability to use several different electron acceptors, including oxygen, nitrate, or organic matter in fermentation. Others, such as methanogens, can only carry out the methane-forming reactions. In fact, methanogens are strict anaerobes, and oxygen is harmful to them.

The above reaction energetics suggest that aerobic organisms need to send relatively few electrons from their donor to oxygen in order to generate the energy required to synthesize a given amount of new biomass. In the terms of our partitioning framework, f_e^0 is small and f_s^0 is large. Since Y and f_s^0 are proportional, the aerobic microorganisms should have a higher true yield (Y) value than anaerobic microorganisms. Thus, knowledge of reaction energetics, as well as energy reactions, gives valuable insight into reaction stoichiometry.

Constructing energy reactions is a useful beginning in the development of an overall stoichiometric equation for microbial growth. Writing equations such as those for glucose oxidation is relatively straightforward. However, more complex reactions, such as those for nitrate and sulfate reduction, are more difficult. Various approaches can be used to do this, including methods described by Battley (1987) and Bailey and Ollis (1986). The approach that we use focuses on half-reactions. The half-reaction approach is the most straightforward to use, particularly for very complex reactions, and it is totally consistent with the energetics of reactions as used here.

The example that follows leads to construction of Equation 2.9, glucose oxidation with nitrate. The oxidation half-reaction for glucose, written on a one-electron-equivalent basis, is

$$\frac{1}{24}\,C_6H_{12}O_6 + \frac{1}{4}\,H_2O \rightarrow \frac{1}{4}\,CO_2 + H^+ + e^- \qquad \textbf{[2.13]}$$

The reduction half-reaction for nitrate, also written for one electron equivalent, is

$$\frac{1}{5}\,NO_3^- + \frac{6}{5}\,H^+ + e^- \rightarrow \frac{1}{10}\,N_2 + \frac{3}{5}\,H_2O \qquad \textbf{[2.14]}$$

Adding Equations 2.13 and 2.14 gives the overall balanced reaction, in which no free electrons (e^-) are present:

$$\frac{1}{24}\,C_6H_{12}O_6 + \frac{1}{5}\,NO_3^- + \frac{1}{5}\,H^+ \rightarrow \frac{1}{4}\,CO_2 + \frac{7}{20}\,H_2O + \frac{1}{10}\,N_2 \qquad \textbf{[2.15]}$$

If Equation 2.15 is multiplied by the least common denominator of 120, Equation 2.9 results.

Tables 2.2 and 2.3 summarize half-reactions, written as one-electron reduction reactions (e^- on the left), for a series of oxidation-reductions of interest in environmental biotechnology. When the half-reaction is to be used as an oxidation, the left and right sides are switched, so that the e^- appears on the right. The sign of the free energy changes when the half-reactions are written as oxidations.

Reduction half-reactions not in the lists can be written readily if a set of simple steps is followed. Consider the reduction half-reaction for the amino acid alanine (CH_3CHNH_2COOH). Only one element is allowed to have its oxidation state changed in a half-reaction. In this case, the element is carbon, which represents 12 e^- eq equivalents in alanine (7 e^- eq in the left-hand CH_3 group, 4 in the middle $CHNH_2$ group, and 1 in the COOH group). The other elements (H, O, and N) must retain their oxidation state the same on both sides of the equation, in this case, +1, −2, and −3, respectively. In half-reactions for organic compounds, the oxidized form is always carbon dioxide, or, in some cases, bicarbonate or carbonate. In the following case, carbon dioxide is used.

Step 1 Write the oxidized form of the element of interest on the left and the reduced form on the right.

$$CO_2 \qquad \rightarrow CH_3CHNH_2COOH$$

Step 2 Add other species that are formed or consumed in the reaction. In oxidation-reduction reactions, water is almost always a reactant or a product; here it will be included as a reactant in order to balance the oxygen present in the organic compound. As a reduction half-reaction, electrons must also appear on the left side of the equation. Since the nitrogen is present in the reduced form as an amino group in alanine, N must appear on the left side of the equation in the reduced form, either as NH_3 or NH_4^+. In this case, we arbitrarily select NH_3 for illustration.

$$CO_2 + H_2O + NH_3 + e^- \qquad \rightarrow CH_3CHNH_2COOH$$

Step 3 Balance the reaction for the element that is reduced and for all elements except oxygen and hydrogen. In this case, carbon and nitrogen must be balanced.

$$3\,CO_2 + H_2O + NH_3 + e^- \qquad \rightarrow CH_3CHNH_2COOH$$

Table 2.2 Inorganic half-reactions and their Gibb's standard free energy at pH = 7.0

Reaction Number	Reduced-oxidized Compounds		Half-reaction	$\Delta G^{0'}$ kJ/e⁻ eq
I-1	Ammonium-Nitrate:	$\frac{1}{8} NO_3^- + \frac{5}{4} H^+ + e^-$	$= \frac{1}{8} NH_4^+ + \frac{3}{8} H_2O$	−35.11
I-2	Ammonium-Nitrite:	$\frac{1}{6} NO_2^- + \frac{4}{3} H^+ + e^-$	$= \frac{1}{6} NH_4^+ + \frac{1}{3} H_2O$	−32.93
I-3	Ammonium-Nitrogen:	$\frac{1}{6} N_2 + \frac{4}{3} H^+ + e^-$	$= \frac{1}{3} NH_4^+$	26.70
I-4	Ferrous-Ferric:	$Fe^{3+} + e^-$	$= Fe^{2+}$	−74.27
I-5	Hydrogen-H⁺:	$H^+ + e^-$	$= \frac{1}{2} H_2$	39.87
I-6	Nitrite-Nitrate:	$\frac{1}{2} NO_3^- + H^+ + e^-$	$= \frac{1}{2} NO_2^- + \frac{1}{2} H_2O$	−41.65
I-7	Nitrogen-Nitrate:	$\frac{1}{5} NO_3^- + \frac{6}{5} H^+ + e^-$	$= \frac{1}{10} N_2 + \frac{3}{5} H_2O$	−72.20
I-8	Nitrogen-Nitrite:	$\frac{1}{3} NO_2^- + \frac{4}{3} H^+ + e^-$	$= \frac{1}{6} N_2 + \frac{2}{3} H_2O$	−92.56
I-9	Sulfide-Sulfate:	$\frac{1}{8} SO_4^{2-} + \frac{19}{16} H^+ + e^-$	$= \frac{1}{16} H_2S + \frac{1}{16} HS^- + \frac{1}{2} H_2O$	20.85
I-10	Sulfide-Sulfite:	$\frac{1}{6} SO_3^{2-} + \frac{5}{4} H^+ + e^-$	$= \frac{1}{12} H_2S + \frac{1}{12} HS^- + \frac{1}{2} H_2O$	11.03
I-11	Sulfite-Sulfate:	$\frac{1}{2} SO_4^{2-} + H^+ + e^-$	$= \frac{1}{2} SO_3^{2-} + \frac{1}{2} H_2O$	50.30
I-12	Sulfur-Sulfate:	$\frac{1}{6} SO_4^{2-} + \frac{4}{3} H^+ + e^-$	$= \frac{1}{6} S + \frac{2}{3} H_2O$	19.15
I-13	Thiosulfate-Sulfate:	$\frac{1}{4} SO_4^{2-} + \frac{5}{4} H^+ + e^-$	$= \frac{1}{8} S_2O_3^{2-} + \frac{5}{8} H_2O$	23.58
I-14	Water-Oxygen:	$\frac{1}{4} O_2 + H^+ + e^-$	$= \frac{1}{2} H_2O$	−78.72

Step 4 Balance the oxygen through addition or subtraction of water. Elemental oxygen is not to be used here, as oxygen must not have its oxidation state changed.

$$3\,CO_2 + NH_3 + e^- \rightarrow CH_3CHNH_2COOH + 4\,H_2O$$

Step 5 Balance hydrogen by introducing H^+.

$$3\,CO_2 + NH_3 + 12\,H^+ + e^- \rightarrow CH_3CHNH_2COOH + 4\,H_2O$$

Table 2.3 Organic half-reactions and their Gibb's free energy

Reaction Number	Reduced Compounds		Half-reaction	$\Delta G^{0'}$ kJ/e⁻ eq
O-1	Acetate:	$\frac{1}{8}CO_2 + \frac{1}{8}HCO_3^- + H^+ + e^-$	$= \frac{1}{8}CH_3COO^- + \frac{3}{8}H_2O$	27.40
O-2	Alanine:	$\frac{1}{6}CO_2 + \frac{1}{12}HCO_3^- + \frac{1}{12}NH_4^+ + \frac{11}{12}H^+ + e^-$	$= \frac{1}{12}CH_3CHNH_2COO^- + \frac{5}{12}H_2O$	31.37
O-3	Benzoate:	$\frac{1}{5}CO_2 + \frac{1}{30}HCO_3^- + H^+ + e^-$	$= \frac{1}{30}C_6H_5COO^- + \frac{13}{30}H_2O$	27.34
O-4	Citrate:	$\frac{1}{6}CO_2 + \frac{1}{6}HCO_3^- + H^+ + e^-$	$= \frac{1}{18}(COO^-)CH_2COH(COO^-)CH_2COO^- + \frac{4}{9}H_2O$	33.08
O-5	Ethanol:	$\frac{1}{6}CO_2 + H^+ + e^-$	$= \frac{1}{12}CH_3CH_2OH + \frac{1}{4}H_2O$	31.18
O-6	Formate:	$\frac{1}{2}HCO_3^- + H^+ + e^-$	$= \frac{1}{2}HCOO^- + \frac{1}{2}H_2O$	39.19
O-7	Glucose:	$\frac{1}{4}CO_2 + H^+ + e^-$	$= \frac{1}{24}C_6H_{12}O_6 + \frac{1}{4}H_2O$	41.35
O-8	Glutamate:	$\frac{1}{6}CO_2 + \frac{1}{9}HCO_3^- + \frac{1}{18}NH_4^+ + H^+ + e^-$	$= \frac{1}{18}COOHCH_2CH_2CHNH_2COO^- + \frac{4}{9}H_2O$	30.93
O-9	Glycerol:	$\frac{3}{14}CO_2 + H^+ + e^-$	$= \frac{1}{14}CH_2OHCHOHCH_2OH + \frac{3}{14}H_2O$	38.88
O-10	Glycine:	$\frac{1}{6}CO_2 + \frac{1}{6}HCO_3^- + \frac{1}{6}NH_4^+ + H^+ + e^-$	$= \frac{1}{6}CH_2NH_2COOH + \frac{1}{2}H_2O$	39.80

O-11	Lactate:	$\frac{1}{6}CO_2 + \frac{1}{12} HCO_3^- + H^+ + e^-$	$= \frac{1}{12} CH_3CHOHCOO^- + \frac{1}{3} H_2O$	32.29
O-12	Methane:	$\frac{1}{8} CO_2 + H^+ + e^-$	$= \frac{1}{8} CH_4 + \frac{1}{4} H_2O$	23.53
O-13	Methanol:	$\frac{1}{6} CO_2 + H^+ + e^-$	$= \frac{1}{6} CH_3OH + \frac{1}{6} H_2O$	36.84
O-14	Palmitate:	$\frac{15}{19} CO_2 + \frac{1}{92} HCO_3^- + H^+ + e^-$	$= \frac{1}{92} CH_3(CH_2)_{14}COO^- + \frac{31}{92} H_2O$	27.26
O-15	Propionate:	$\frac{1}{7} CO_2 + \frac{1}{14} HCO_3^- + H^+ + e^-$	$= \frac{1}{14} CH_3CH_2COO^- + \frac{5}{14} H_2O$	27.63
O-16	Pyruvate:	$\frac{1}{5} CO_2 + \frac{1}{10} HCO_3^- + H^+ + e^-$	$= \frac{1}{10} CH_3COCOO^- + \frac{2}{5} H_2O$	35.09
O-17	Succinate:	$\frac{1}{7} CO_2 + \frac{1}{7} HCO_3^- + H^+ + e^-$	$= \frac{1}{14} (CH_2)_2(COO^-)_2 + \frac{3}{7} H_2O$	29.09
O-18	Domestic Wastewater:	$\frac{9}{50} CO_2 + \frac{1}{50} NH_4^+ + \frac{1}{50} HCO_3^- + H^+ + e^-$	$= \frac{1}{50} C_{10}H_{19}O_3N + \frac{9}{25} H_2O$	*
O-19	Custom Organic Half Reaction:	$\frac{(n-c)}{d} CO_2 + \frac{c}{d} NH_4^+ + \frac{c}{d} HCO_3^- + H^+ + e^-$	$= \frac{1}{d} C_nH_aO_bN_c + \frac{2n-b+c}{d} H_2O$ where, $d = (4n + a - 2b - 3c)$	*
O-20	Cell Synthesis:	$\frac{1}{5} CO_2 + \frac{1}{20} NH_4^+ + \frac{1}{20} HCO_3^- + H^+ + e^-$	$= \frac{1}{20} C_5H_7O_2N + \frac{9}{20} H_2O$	*

*Equations O-18 to O-20 do not have $\Delta G^{0'}$ values because the reduced species is not chemically defined.

Step 6 Balance the charge on the reaction by adding sufficient e^- to the left side of the equation.

$$3\,CO_2 + NH_3 + 12\,H^+ + 12\,e^- \rightarrow CH_3CHNH_2COOH + 4\,H_2O$$

The coefficient on electrons should equal the number of electron equivalents in the reduced compound, or 12 e^- eq in alanine, which it does.

Step 7 Convert the equation to the electron-equivalent form by dividing by the coefficient on e^-.

$$\frac{1}{4}\,CO_2 + \frac{1}{12}\,NH_3 + H^+ + e^- \rightarrow \frac{1}{12}\,CH_3CHNH_2COOH + \frac{1}{3}\,H_2O \qquad \textbf{[2.16]}$$

Inorganic oxidation-reduction half-reactions can be written in a similar fashion. The reduced and oxidized form of the element must be known. For example, chromium commonly exists in aqueous solution in either the oxidized Cr(VI) form or the reduced Cr(III) form. The chemical species associated with Cr(VI) in water are the oxyanion forms, either CrO_4^{2-} at neutral pH, or as the dichromate form, $Cr_2O_7^{2-}$, in acid waters. Cr(III) would exist in water as a cation, Cr^{3+}. Assuming that neutral-pH conditions apply and following the above seven steps, we derive the reduction half-reaction for chromate by following the seven steps.

Step 1

$$CrO_4^{2-} \rightarrow Cr^{3+}$$

Step 2

$$CrO_4^{2-} + H^+ + e^- \rightarrow Cr^{3+} + H_2O$$

Step 3

$$CrO_4^{2-} + H^+ + e^- \rightarrow Cr^{3+} + H_2O \quad \text{(Cr balanced in Step 2; thus no change here)}$$

Step 4

$$CrO_4^{2-} + H^+ + e^- \rightarrow Cr^{3+} + 4\,H_2O$$

Step 5

$$CrO_4^{2-} + 8\,H^+ + e^- \rightarrow Cr^{3+} + 4\,H_2O$$

Step 6

$$CrO_4^{2-} + 8\,H^+ + 3\,e^- \rightarrow Cr^{3+} + 4\,H_2O$$

Step 7

$$\frac{1}{3}\,CrO_4^{2-} + \frac{8}{3}\,H^+ + e^- \rightarrow \frac{1}{3}\,Cr^{3+} + \frac{4}{3}\,H_2O$$

One half-reaction of note is the last one in Table 2.3. It is for the formation of bacterial cells based on the empirical formula $C_5H_7O_2N$:

$$\frac{1}{5}\,CO_2 + \frac{1}{20}\,NH_4^+ + \frac{1}{20}\,HCO_3^- + H^+ + e^- \rightarrow \frac{1}{20}\,C_5H_7O_2N + \frac{9}{20}\,H_2O \qquad [2.17]$$

Also listed in Table 2.3 is a generic equation for organic compounds in general (reaction O-19) based on an assumed formula for organic matter of $C_nH_aO_bN_c$. Organic compounds may contain other elements, such as P and S, and these may be added if desired. Also, organic compounds may have charge, such as negative charge for carboxylate anions or positive charge for some nitrogen-containing organic compounds. Charge could be added as well if needed. Another representation in Table 2.3 is an empirical formula for organic matter in domestic sewage (reaction O-18). Such representations can be developed if knowledge is available on the relative composition of a wastewater in terms of carbon, oxygen, hydrogen, and nitrogen in the organic matter.

If Equation 2.16 is compared with its counterpart in Table 2.3 (reaction O-2), there are differences, particularly in the distribution of CO_2, HCO_3^-, H_2O, and H^+. Yet, the number of moles of alanine given by one electron equivalent (1/12) remains the same. Both equations balance. Thus, questions may be asked, such as: Why the difference? Which equation is correct? Which one should I use for my particular problem? One answer is that both are correct from the standpoint of being balanced half-reactions. The difference results from the assumed species that are involved in the reaction.

For example, in one equation, NH_3 is assumed to be the species of nitrogen involved, while in the other, the ionized form, NH_4^+, is assumed. The best form to include depends on the problem being addressed. If we are describing the biodegradation of alanine in a biological treatment reactor operating at neutral pH, then an assumption that NH_4^+ is formed in the reaction is perhaps the better one, since this is the species that dominates in aqueous solutions at pH below about 9.3. If the interest were in a reaction occurring at pH well above this value, then NH_3 would be the species expected to dominate. When NH_4^+ is released following oxidation of an organic compound, a negatively charged species must also be formed to balance the charge. With organic oxidation at neutral pH, the species thus formed is generally HCO_3^-. It is for this reason that, for each half-reaction in Table 2.3 that involves organic nitrogen, HCO_3^- enters into the reaction. The significance of this may be illustrated by considering the overall oxidation-reduction reaction that occurs in the anaerobic methanogenic fermentation of alanine, which can be obtained by subtracting reaction O-2 from reaction O-12:

$$\text{Reaction O-12} \quad \frac{1}{8}\,CO_2 + H^+ + e^- \rightarrow \frac{1}{8}\,CH_4 + \frac{1}{4}\,H_2O \qquad [2.18]$$

$$-\text{Reaction O-2} \quad \frac{1}{12}\,CH_3CHNH_2COOH + \frac{5}{12}\,H_2O \rightarrow \frac{1}{6}\,CO_2 + \frac{1}{12}\,NH_4^+ + \frac{1}{12}\,HCO_3^- + H^+ + e^- \qquad [2.19]$$

$$\text{Sum} \quad \frac{1}{12}\,CH_3CHNH_2COOH + \frac{1}{6}\,H_2O \rightarrow \frac{1}{8}\,CH_4 + \frac{1}{24}\,CO_2 + \frac{1}{12}\,NH_4^+ + \frac{1}{12}\,HCO_3^- \qquad [2.20]$$

This equation indicates that, when 1/12 mole (or 7.42 g) of alanine is converted for energy during methane fermentation, 1/8 mole methane and 1/24 mole carbon dioxide are formed as gases, and 1/12 mole each of ammonium and bicarbonate are released to the solution. This correlation between NH_4^+ and HCO_3^- is very important for pH control in anaerobic treatment and is emphasized again in Chapter 13.

Example 2.3 **PRODUCTS FROM METHANOGENESIS OF ALANINE** An industrial wastewater contains 100-mM alanine. If it were all converted for energy during methane fermentation at a pH near neutral: (a) what would be the composition of the gas produced, and (b) what would be the concentrations of ammonium and bicarbonate in solution?

(a) Since the gas produced has three times more methane than carbon dioxide, according to Equation 2.20, then the composition of the gases produced is 75% methane and 25% carbon dioxide.

(b) Since Equation 2.20 indicates that 1/12 mole each of NH_4^+ and HCO_3^- are produced for each 1/12 mole of alanine fermented to methane for energy, then 100-mM alanine produces 100 mM each of ammonia nitrogen and bicarbonate. With a formula weight of 14, the ammonium-nitrogen concentration would increase by 1,400 mg/l. Also, the bicarbonate alkalinity, when expressed as equivalent calcium carbonate (equivalent weight of 50), increases by 5,000 mg/l. The bicarbonate formed produces a natural pH buffer to help maintain a near-neutral pH.

For half reactions that involve organic chemicals with a negative charge, such as the carboxylate group for acetic acid, HCO_3^- can be used in the reaction to provide charge balance. Several examples are contained in Table 2.3; that for pyruvate is repeated here as an example:

$$\frac{1}{5}CO_2 + \frac{1}{10}HCO_3^- + H^+ + e^- \rightarrow \frac{1}{10}CH_3COCOO^- + \frac{2}{5}H_2O \qquad [2.21]$$

Here, the moles of bicarbonate on the left side of the equation are made to just equal the moles of carboxylate on the right side. Carbon dioxide is then added in sufficient amount to provide a carbon balance. We use HCO_3^- when NH_4^+ nitrogen or organic anions are involved because the species conform to the dominant species in solution at neutral pH. Some authors prefer other forms for different reasons. All are technically correct as long as the equations are written in a balanced form. Different half-reaction forms sometimes used are illustrated in the following half-reactions for alanine:

$$\frac{1}{4}CO_2 + \frac{1}{12}NH_3 + H^+ + e^- \rightarrow \frac{1}{12}CH_3CHNH_2COOH + \frac{1}{3}H_2O \qquad [2.22]$$

$$\frac{1}{6}CO_2 + \frac{1}{12}NH_4^+ + \frac{1}{12}HCO_3^- + H^+ + e^- \rightarrow \frac{1}{12}CH_3CHNH_2COOH + \frac{5}{12}H_2O \qquad [2.23]$$

$$\frac{1}{6}CO_2 + \frac{1}{12}NH_4^+ + \frac{1}{12}HCO_3^- + \frac{11}{12}H^+ + e^- \rightarrow \frac{1}{12}CH_3CHNH_2COO^- + \frac{5}{12}H_2O \qquad [2.23a]$$

$$\frac{1}{6}CO_2 + \frac{1}{12}NH_4^+ + \frac{1}{12}HCO_3^- + \frac{1}{2}H_2 \rightarrow \frac{1}{12}CH_3CHNH_2COOH + \frac{5}{12}H_2O \qquad [2.24]$$

$$\frac{1}{12}NH_4^+ + \frac{1}{4}HCO_3^- + \frac{13}{12}H^+ + e^- \rightarrow \frac{1}{12}CH_3CHNH_2COO^- + \frac{7}{12}H_2O \qquad [2.25]$$

Still other forms can be written through various combinations of the above. Regardless of the form selected, 1/12 mole of alanine gives one electron equivalent.

2.5 OVERALL REACTIONS FOR BIOLOGICAL GROWTH

Bacterial growth involves two basic reactions, one for energy production and the other for cellular synthesis. The electron donor provides electrons to the electron acceptor for energy production. The previous section showed how to combine half-reactions for the energy part. The next step is to combine half-reactions to represent *synthesis*. With this information and knowledge of how the electrons from the donor are partitioned, we can write an overall, balanced reaction such as Equation 2.1.

Before we can do this, we need to have a half-reaction for synthesis, R_c. Table 2.4 lists the key synthesis half-reactions. We now consider the first synthesis half-reaction (C-1), which is used when ammonium is the nitrogen source for formation of proteins and nucleic acids. Ammonium is the preferred nitrogen source. If it is not available, microorganisms may be able to use the other nitrogen sources indicated in the table. Also contained in Table 2.4 for convenience are the acceptor half-reactions (R_a) for the five most common electron acceptors: O_2, NO_3^-, Fe^{3+}, SO_4^{2-}, and CO_2.

As an example, we assume that benzoate is available as an electron donor, nitrate is available as an electron acceptor, and ammonium is available as the nitrogen source. On a net-yield basis, we also assume that 40 percent of the electron equivalents in benzoate is used for synthesis ($f_s = 0.40$), while the other 60 percent is used for energy ($f_e = 0.60$).

First, the overall energy and synthesis reactions are developed. The donor half-reaction is designated as R_d, the acceptor half-reaction as R_a, and the cell half-reaction as R_c. The energy reaction, R_e, then becomes

$$R_e = R_a - R_d \qquad \textbf{[2.26]}$$

and the synthesis reaction R_s, becomes

$$R_s = R_c - R_d \qquad \textbf{[2.27]}$$

It is important to note that R_d has a negative sign, because the donor is oxidized.

The actual half-reactions are now substituted into Equations 2.26 and 2.27. First is the energy reaction:

$$R_a:\quad \frac{1}{5}\,NO_3^- + \frac{6}{5}\,H^+ + e^- \rightarrow \frac{1}{10}\,N_2 + \frac{3}{5}\,H_2O \qquad \textbf{[2.28]}$$

$$-R_d:\quad \frac{1}{30}\,C_6H_5COO^- + \frac{13}{30}\,H_2O \rightarrow \frac{1}{5}\,CO_2 + \frac{1}{30}\,HCO_3^- + H^+ + e^- \qquad \textbf{[2.29]}$$

$$R_e:\quad \frac{1}{30}\,C_6H_5COO^- + \frac{1}{5}\,NO_3^- + \frac{1}{5}\,H^+ \rightarrow \frac{1}{5}\,CO_2 + \frac{1}{10}\,N_2 + \frac{1}{30}\,HCO_3^- + \frac{1}{6}\,H_2O \qquad \textbf{[2.30]}$$

Table 2.4 Cell formation (R_c) and common electron acceptor half-reactions (R_a)

Reaction Number		Half-reaction		$\Delta G^{0'}$ kJ/e^- eq
Cell Synthesis Equations (R_c)				
Ammonium as Nitrogen Source				
C-1	$\frac{1}{5} CO_2 + \frac{1}{20} HCO_3^- + \frac{1}{20} NH_4^+ + H^+ + e^-$	$= \frac{1}{20} C_5H_7O_2N + \frac{9}{20} H_2O$		
Nitrate as Nitrogen Source				
C-2	$\frac{1}{28} NO_3^- + \frac{5}{28} CO_2 + \frac{29}{28} H^+ + e^-$	$= \frac{1}{28} C_5H_7O_2N + \frac{11}{28} H_2O$		
Nitrite as Nitrogen Source				
C-3	$\frac{5}{26} CO_2 + \frac{1}{26} NO_2^- + \frac{27}{26} H^+ + e^-$	$= \frac{1}{26} C_5H_7O_2N + \frac{10}{26} H_2O$		
Dinitrogen as Nitrogen Source				
C-4	$\frac{5}{23} CO_2 + \frac{1}{46} N_2 + H^+ + e^-$	$= \frac{1}{23} C_5H_7O_2N + \frac{8}{23} H_2O$		
Common Electron-Acceptor Equations (R_a)				
I-14 Oxygen	$\frac{1}{4} O_2 + H^+ + e^-$	$= \frac{1}{2} H_2O$		−78.72
I-7 Nitrate	$\frac{1}{5} NO_3^- + \frac{6}{5} H^+ + e^-$	$= \frac{1}{10} N_2 + \frac{3}{5} H_2O$		−72.20
I-9 Sulfate	$\frac{1}{8} SO_4^{2-} + \frac{19}{16} H^+ + e^-$	$= \frac{1}{16} H_2S + \frac{1}{16} HS^- + \frac{1}{2} H_2O$		20.85
O-12 CO_2	$\frac{1}{8} CO_2 + H^+ + e^-$	$= \frac{1}{8} CH_4 + \frac{1}{4} H_2O$		23.53
I-4 Iron (III)	$Fe^{3+} + e^-$	$= Fe^{2+}$		−74.27

Similarly, the synthesis reaction is

$$R_c: \quad \frac{1}{5} CO_2 + \frac{1}{20} NH_4^+ + \frac{1}{20} HCO_3^- + H^+ + e^- \rightarrow \frac{1}{20} C_5H_7O_2N + \frac{9}{20} H_20 \qquad [2.17]$$

$$-R_d: \quad \frac{1}{30} C_6H_5COO^- + \frac{13}{30} H_2O \rightarrow \frac{1}{5} CO_2 + \frac{1}{30} HCO_3^- + H^+ + e^- \qquad [2.29]$$

$$R_s: \quad \frac{1}{30} C_6H_5COO^- + \frac{1}{20} NH_4^+ + \frac{1}{60} HCO_3^- \rightarrow \frac{1}{20} C_5H_7O_2N + \frac{1}{60} H_2O \qquad [2.31]$$

Second, and in order to obtain an overall reaction that includes energy generation and synthesis, Equation 2.30 is multiplied by f_e, Equation 2.31 is multiplied by f_s, and they are summed:

$$f_eR_e: \quad 0.02\ C_6H_5COO^- + 0.12\ NO_3^- + 0.12\ H^+ \rightarrow 0.12\ CO_2 + 0.06\ N_2 + 0.02\ HCO_3{}^- + 0.1\ H_2O \qquad [2.32]$$

$$f_sR_s: \quad 0.0133\ C_6H_5COO^- + 0.02\ NH_4^+ + 0.0067\ HCO_3^- \rightarrow 0.02\ C_5H_7O_2N + 0.0067\ H_2O \qquad [2.33]$$

$$R: \quad 0.0333\ C_6H_5COO^- + 0.12\ NO_3^- + 0.02\ NH_4^+ + 0.12\ H^+ \rightarrow 0.02\ C_5H_7O_2N + 0.06\ N_2 + 0.12\ CO_2 + 0.0133\ HCO_3^- + 0.1067\ H_2O \qquad [2.34]$$

Equation 2.34 provides an overall equation for net synthesis of bacteria that are using benzoate as an electron donor and nitrate as an electron acceptor. The bacteria here also use ammonium as a nitrogen source for synthesis.

We can see that Equation 2.34 is obtained by summation of Equation 2.32 and 2.33, that is,

$$R = f_e(R_a - R_d) + f_s(R_c - R_d) \qquad \textbf{[2.35]}$$

Since the fractions of electrons used for energy generation and synthesis must equal 1.0,

$$f_s + f_e = 1.0 \quad \text{and} \quad R_d(f_s + f_e) = R_d \qquad \textbf{[2.36]}$$

which converts Equation 2.35 to

$$R = f_e R_a + f_s R_c - R_d \qquad \textbf{[2.37]}$$

Equation 2.37 is a general equation that can be used for constructing a wide variety of stoichiometric equations for microbial synthesis and growth. The result is an equation written on an electron-equivalent basis. In other words, the equation represents the net consumption of reactants and production of products when the microorganisms consume one electron equivalent of electron donor.

The bacteria in the above case with benzoate used ammonium as a nitrogen source for synthesis. In some situations, ammonium is not available, but oxidized sources of nitrogen are available. The oxidized sources are NO_3^-, NO_2^-, and N_2, although NO_3^- is the most common one. Table 2.4 gives the cell-formation half-reactions for all four nitrogen sources. An important feature of the half-reactions is that the cell formula has a different stoichiometric coefficient for each nitrogen source: 1/20 for NH_4^+, 1/28 for NO_3^-, 1/26 for NO_2^-, and 1/23 for N_2. This difference reflects the number of electron equivalents that must be added to the 20 for C when the N source for $C_5H_7O_2N$ must be reduced: 0 extra for NH_4^+, 8 extra for NO_3^-, 6 extra for NO_2^-, and 3 extra for N_2. When an oxidized N source is required, the f_s value also may change.

The use of an oxidized nitrogen source is illustrated here for NO_3^-. We write an overall equation for bacterial growth with benzoate as the electron donor and carbon source, while NO_3^- is the electron acceptor and nitrogen source. For this case, we assume that f_s is 0.55. The appropriate equations are O-3 from Table 2.3 for benzoate and R_d, C-2 from Table 2.4 for R_c, and I-7 from Table 2.4 for R_a. The value of f_e becomes $1 - f_s = 0.45$. The overall reaction is then determined by the same pattern:

$$f_e R_a\text{:} \quad 0.09\ NO_3^- + 0.54\ H^+ + 0.45\ e^- \rightarrow 0.045\ N_2 + 0.27\ H_2O$$

$$f_s R_c\text{:} \quad 0.0196\ NO_3^- + 0.0982\ CO_2 + 0.5696\ H^+ + 0.55\ e^- \rightarrow 0.0196\ C_5H_7O_2N + 0.2161\ H_2O$$

$$-R_d\text{:} \quad 0.0333\ C_6H_5COO^- + 0.4333\ H_2O \rightarrow 0.2\ CO_2 + 0.0333\ HCO_3^- + H^+ + e^-$$

$$R\text{:} \quad 0.0333\ C_6H_5COO^- + 0.1096\ NO_3^- + 0.1096\ H^+ \rightarrow$$
$$0.0196\ C_5H_7O_2N + 0.045\ N_2 + 0.0333\ HCO_3^- + 0.1018\ CO_2 + 0.0528\ H_2O$$

We see that 0.09 mole nitrate is converted to nitrogen gas, and 0.0196 mole is converted into the organic nitrogen of the cells.

Most of the examples provided up to this point are for oxidations of organic matter. However, chemolithotrophs are important microorganisms that use reduced inorganic chemicals for energy and most frequently use inorganic carbon (CO_2) for organism synthesis. The inorganic half-reactions listed in Table 2.2 are examples from the many different inorganic oxidation-reduction reactions that can be mediated by lithotrophic microorganisms in order to obtain energy for growth. An important aspect to be noted from Equation 2.37 and its development is that the electron equivalents from the electron donor are divided between the energy reaction and cell synthesis. While the preceding examples illustrate this for heterotrophic reactions, it is equally true for lithotrophic reactions.

Example 2.4

NITRIFICATION STOICHIOMETRY Lithotrophic microorganisms are employed to oxidize ammonium in a wastewater to nitrate under aerobic conditions in order to reduce the oxygen consumption by nitrification in a receiving stream. If the concentration of ammonium in the wastewater, expressed as nitrogen, is 22 mg/l, how much oxygen will be consumed for nitrification in the treatment of 1,000 m^3 of wastewater, what mass of cells in kg dry weight will be produced, and what will be the resulting concentration of nitrate-nitrogen in the treated water? Assume f_s equals 0.10 and that inorganic carbon is used for cell synthesis.

For the reaction involved, ammonium serves as the electron donor and is oxidized to nitrate, oxygen is the electron acceptor since it is an aerobic reaction, and ammonium likewise serves as the source of nitrogen for cell synthesis. In addition, $f_e = 1 - f_s = 0.90$. Selecting the appropriate half-reactions from Tables 2.2 and 2.4 and employing Equation 2.37 gives the following overall biological reaction:

$$f_e R_a\text{:}\quad 0.225\ O_2 + 0.9\ H^+ + 0.9\ e^- \rightarrow 0.45\ H_2O$$

$$f_s R_c\text{:}\quad 0.02\ CO_2 + 0.005\ NH_4^+ + 0.005\ HCO_3^- + 0.1\ H^+ + 0.1\ e^- \rightarrow 0.005\ C_5H_7O_2N + 0.045\ H_2O$$

$$-R_d\text{:}\quad 0.125\ NH_4^+ + 0.375\ H_2O \rightarrow 0.125\ NO_3^- + 1.25\ H^+ + e^-$$

$$R\text{:}\quad 0.13\ NH_4^+ + 0.225\ O_2 + 0.02\ CO_2 + 0.005\ HCO_3^- \rightarrow 0.005\ C_5H_7O_2N + 0.125\ NO_3^- + 0.25\ H^+ + 0.12\ H_2O$$

For each 0.13(14) = 1.82 g ammonium-nitrogen, 0.225(32) = 7.2 g oxygen is consumed. Also, 0.005(113) = 0.565 g cells and 0.125(14) = 1.75 g NO_3^--N are produced. The amount of ammonium-nitrogen treated = (22 mg/l)(1000 m^3)(10^3 liters/m^3)(kg/10^6 mg) = 22 kg. Thus,

$$\begin{aligned} \text{Oxygen consumption} &= 22\text{ kg}(7.2\text{ g}/1.82\text{ g}) &&= 87\text{ kg} \\ \text{Cell dry weight produced} &= 22\text{ kg}(0.565\text{ g}/1.82\text{ g}) &&= 6.83\text{ kg} \\ \text{Effluent } NO_3^-\text{-N conc.} &= 22\text{ mg/l}(1.75\text{ g}/1.82\text{ g}) &&= 21\text{ mg/l} \end{aligned}$$

Example 2.5

METHANOGENESIS STOICHIOMETRY Based upon analysis of its organic carbon, hydrogen, oxygen, and nitrogen content, the representative empirical formula of $C_8H_{17}O_3N$ was developed for organic matter in an industrial wastewater, and, from the same analysis, the organic concentration was found to be 23,000 mg/l. For a wastewater flow of 150 m^3/d, what is the daily methane production from the wastewater in m^3 at 35 °C and 1 atm and methane percentage in the gas produced if the waste were treated anaerobically by methane fermentation? Assume f_s is 0.08, the process is 95 percent efficient at removal of organic matter, and all gases formed evolve to the gas phase.

First, an electron-donor half-reaction for the wastewater needs to be developed, Applying Equation O-19 of Table 2.3 and the empirical formula ($C_8H_{17}O_3N$) give R_d:

$$R_d: \quad \frac{1}{40}\, NH_4^+ + \frac{1}{40}\, HCO_3^- + \frac{7}{40}\, CO_2 + H^+ + e^- \rightarrow \frac{1}{40}\, C_8H_{17}O_3N + \frac{7}{20}\, H_2O$$

The quantity of organic matter removed per day = 0.95(23 kg/m^3)(150 m^3/d) = 3,280 kg/d.

Using $f_e = 1 - f_s = 1 - 0.08 = 0.92$ and the half-reaction for CO_2 to CH_4 as the electron acceptor reaction O-12 from Table 2.3, we develop R:

$$f_eR_a: \; 0.115\, CO_2 + 0.92\, H^+ + 0.92\, e^- \rightarrow 0.115\, CH_4 + 0.23\, H_20$$

$$f_sR_c: \; 0.016\, CO_2 + 0.004\, NH_4^+ + 0.004\, HCO_3^- + 0.08\, H^+ + 0.08\, e^- \rightarrow 0.004\, C_5H_7O_2N + 0.036\, H_20$$

$$-R_d: \; 0.025\, C_8H_{17}O_3N + 0.35\, H_2O \rightarrow 0.025\, NH_4^+ + 0.025\, HCO_3^- + 0.175\, CO_2 + H^+ + e^-$$

$$R: \; 0.025\, C_8H_{17}O_3N + 0.084\, H_2O \rightarrow 0.004\, C_5H_7O_2N + 0.115\, CH_4 + 0.044\, CO_2 + 0.021\, NH_4^+ + 0.021\, HCO_3^-$$

The formula weight of $C_8H_{17}O_3N$ is 175, and the equivalent weight is 0.025(175), or 4.375 g. Methane fermentation of one equivalent of organic matter produces 0.115 mol methane and 0.044 mol carbon dioxide. Thus,

$$\text{Methane produced} = [(273+35)/273][0.0224\text{m}^3\text{ gas/mol}][3{,}280{,}000\text{g/d}][0.115\text{ mol}/4.375\text{ g}]$$
$$= 2{,}180\text{ m}^3/\text{d}$$

$$\text{Percent methane} = 100[0.115/(0.115+0.044)] = 72\text{ percent}$$

2.5.1 FERMENTATION REACTIONS

In fermentations, an organic compound serves as electron donor and electron acceptor. A simple example is ethanol fermentation from glucose, as given by Equation 2.12. Here, one mole of glucose is converted into two moles ethanol and two moles carbon dioxide. We would like to develop a procedure for writing such an equation that is compatible with Equation 2.37, which we have been using to write overall reactions for biological growth. To do this, we again rely on half-reactions. We need to select appropriate half-reactions for electron donors and electron acceptors.

The complex pathways by which glucose is converted into ethanol were described in Chapter 1. From the viewpoint of writing an overall balanced reaction resulting from this process, one need not consider all the intermediate processes involved. No matter how complicated are the pathways by which chemical A is transformed into chemical B, the laws of energy and mass conservation must be obeyed. Knowledge of all reactants and products in a given case is all that is required to construct a balanced equation for the reaction, and we do not need to know the intermediates along the path, as long as they do not persist.

Simple Fermentations A simple fermentation has only one reduced product, such as ethanol from glucose. All the electrons starting in glucose must end up in ethanol. Our first task is to select an electron donor half-reaction to use in Equation 2.37. The donor obviously is glucose, and the half-reaction we will use is that from Table 2.3 for conversion of CO_2 to glucose (reaction O-7). The second task is to identify the electron-acceptor reaction. It also is rather simple: the half-reaction for CO_2 to ethanol from Table 2.3 (reaction O-5). For the energy reaction alone, we simply use Equation 2.26:

$$R_a: \quad \frac{1}{6} CO_2 + H^+ + e^- \rightarrow \frac{1}{12} CH_3CH_2OH + \frac{1}{4} H_2O$$

$$-R_d: \quad \frac{1}{24} C_6H_{12}O_6 + \frac{1}{4} H_2O \rightarrow \frac{1}{4} CO_2 + H^+ + e^-$$

$$R_e: \quad \frac{1}{24} C_6H_{12}O_6 \rightarrow \frac{1}{12} CH_3CH_2OH + \frac{1}{12} CO_2$$

Multiplying R_e by 24 equivalents per mole gives Equation 2.12. Also, the selection of electron donor and acceptor half-reactions in this manner is compatible with using Equation 2.37 to build an overall reaction.

Example 2.6 **SIMPLE FERMENTATION STOICHIOMETRY** Write the overall biological reaction for ethanol fermentation from glucose, assuming that f_s equals 0.22.

Using the preceding ethanol and glucose half-reactions, coupled with the cell synthesis reaction from Table 2.3, and $f_e = 1 - f_s = 0.78$:

$$0.78R_a: \; 0.13\ CO_2 + 0.78\ H^+ + 0.78\ e^- \rightarrow 0.065\ CH_3CH_2OH + 0.195\ H_2O$$

$$0.22R_c: \; 0.044\ CO_2 + 0.011\ NH_4^+ + 0.011\ HCO_3^- + 0.22\ H^+ + 0.22\ e^- \rightarrow 0.011\ C_5H_7O_2N + 0.099\ H_2O$$

$$-R_d: \; 0.0417\ C_6H_{12}O_6 + 0.25\ H_2O \rightarrow 0.25\ CO_2 + H^+ + e^-$$

$$R: \quad 0.0417\ C_6H_{12}O_6 + 0.011\ NH_4^+ + 0.011\ HCO_3^- \rightarrow 0.011\ C_5H_7O_2N + 0.065\ CH_3CH_2OH + 0.076\ CO_2 + 0.044\ H_2O$$

This reaction indicates that for every equivalent of glucose fermented, 0.065 mol ethanol is formed. Also, 0.011 mol ammonium is required to produce 0.011 empirical mol of microbial cells. Carbon dioxide is also produced in the process, which, if the fermentation is carried out in a closed bottle, will provide carbonation, if that is desired!

Mixed Fermentations Many fermentations form more than a single product. For example, *E. coli* generally ferments glucose to a mixture of acetate, ethanol, formate, and hydrogen. In methane fermentation, a consortium of Bacteria and Archaea converts organic material into methane, and the products of incomplete fermentation, such as acetate, propionate, and butyrate, often are present. Once the relative proportions of reduced end products are known, the energy reaction can be constructed. The reduced products include all organic products plus hydrogen (H_2). The carbon dioxide formed is not as important in analyzing fermentations, since it is fully oxidized.

The critical step is determining the relative proportion of electron equivalents represented by each of the reduced end products. The electron equivalents of each product are computed, the sum is taken, and the fraction that each reduced end product represents of the total is then computed. That fraction is then used as a multiplier for the half-reaction for each reduced product, and the resulting equations are added together to produce a half-reaction for the electron acceptor, R_a. This series of operations is written in mathematical form as

$$R_a = \sum_{i=1}^{n} e_{ai} R_{ai} \qquad \textbf{[2.38]}$$

where

$$e_{ai} = \frac{\text{equiv}_{ai}}{\sum_{j=1}^{n} \text{equiv}_{aj}} \quad \text{and} \quad \sum_{i=1}^{n} e_{ai} = 1$$

Here, e_{ai} is the fraction of the n reduced end products that is represented by product a_i. Equiv$_{ai}$ represents the equivalents of a_i produced. The sum of the fractions of all reduced end products equals 1.

Some cases could have mixed electron donors. Certainly this is the case in the treatment of municipal and industrial wastewaters. Here, the electron-donor reaction, R_d, is written similar to that above for the electron acceptor:

$$R_d = \sum_{i=1}^{n} e_{di} R_{di} \qquad \textbf{[2.39]}$$

where

$$e_{di} = \frac{\text{equiv}_{di}}{\sum_{j=1}^{n} \text{equiv}_{dj}} \quad \text{and} \quad \sum_{i=1}^{n} e_{di} = 1$$

Example 2.7

CITRATE FERMENTATION TO TWO REDUCED PRODUCTS *Bacteroides* sp. converts 1 mol citrate into 1 mol formate, 2 mol acetate, and 1 mol bicarbonate. Write the overall balanced energy reaction (R_e) for this fermentation.

The reduced end products are formate and acetate. Bicarbonate, like carbon dioxide, is an oxidized end product and not considered in constructing the electron balance. The first step is to determine the number of equivalents (equiv$_{ai}$) formed for each reduced product. From reaction O-1, Table 2.3, there are 8 e^- eq in a mole of acetate; so 2 mol acetate represents 16 e^- eq. Similarly, there are 2 e^- eq of formate per mole. Then,

$$e_{\text{formate}} = 2/(2+16) \text{ or } 0.111, \text{ and } e_{\text{acetate}} = 16/(2+16) \text{ or } 0.889.$$

The sum of e_{formate} plus e_{acetate} equals 1.0.

Using half-reactions from Table 2.3:

$$0.111\ R_{formate}:\quad 0.0555\ HCO_3^- + 0.111\ H^+ + 0.111\ e^- \rightarrow 0.0555\ HCOO^- + 0.0555\ H_2O$$

$$0.889\ R_{acetate}:\quad 0.111\ CO_2 + 0.111\ HCO_3^- + 0.889\ H^+ + 0.889\ e^- \rightarrow 0.111\ CH_3COO^- + 0.333\ H_2O$$

$$R_a:\quad 0.111\ CO_2 + 0.166\ HCO_3^- + H^+ + e^- \rightarrow 0.0555\ HCOO^- + 0.111\ CH_3COO^- + 0.388\ H_2O$$

The overall energy reaction (R_e) is then found using Equation 2.37, or $R_a - R_d$, where R_d represents the half-reaction for citrate from Table 2.3. Combining these produces the following for R_e:

$$0.0555\ (COO^-)CH_2COH(COO^-)CH_2COO^- + 0.056\ H_2O \rightarrow$$
$$0.0555\ HCOO^- + 0.111\ CH_3COO^- + 0.056\ CO_2$$

If this equation is normalized by dividing through by 0.0555, the moles of citrate in one equivalent, the following mole-normalized equation is obtained:

$$(COO^-)CH_2COH(COO^-)CH_2COO^- + H_2O \rightarrow HCOO^- + 2\ CH_3COO^- + CO_2$$

It is readily apparent that the equation satisfies the requirement that 1 mol formate and 2 mol acetate are formed from 1 mol citrate. This is a fairly simple example that could have been balanced by other procedures. However, the next example provides a more complicated case, one of mixed reactants and products.

Example 2.8

FERMENTATION WITH MIXED DONORS AND PRODUCTS In the methane fermentation of a mixture containing 1.1 mol lactate and 1.1 mol glucose, the distribution of reduced end products was: 3.6 mol methane, 0.21 mol acetate, and 0.42 mol propionate. Write a balanced energy reaction for this case.

Based upon the electron equivalents noted in Table 2.3, the following table can be constructed:

Electron-Donor Substrates	**Moles**	**e^- eq/mol**	**Equiv$_{di}$**	**e_{di}**
lactate	1.1	12	$1.1 \times 12 = 13.2$	$13.2/39.6 = 0.33$
glucose	1.1	24	$1.1 \times 24 = 26.4$	$26.4/39.6 = 0.67$
			$\sum = 39.6$	$\sum = 1.00$
Electron Acceptor Products	**Moles**	**e^- eq/mol**	**Equiv$_{ai}$**	**e_{ai}**
methane	3.6	8	$3.6 \times 8 = 28.8$	$28.8/36.22 = 0.796$
acetate	0.21	8	$0.21 \times 8 = 1.68$	$1.68/36.22 = 0.046$
propionate	0.41	14	$0.41 \times 14 = 5.74$	$5.74/36.22 = 0.158$
			$\sum = 36.22$	$\sum = 1.000$

The sum of the electron equivalents for the donor is about 10 percent larger than the sum of the equivalents for the acceptor. Part of the donor in a biological reaction is associated with cell synthesis; when net synthesis occurs, the donor equivalents should be larger, as is the case here.

The acceptor reaction is then constructed first:

$$0.796R_{\text{meth}}:\ 0.0995\ CO_2 + 0.796\ H^+ + 0.796\ e^- \rightarrow 0.0995\ CH_4 + 0.199\ H_2O$$

$$0.046R_{\text{acet}}:\ 0.0058\ CO_2 + 0.0058\ HCO_3^- + 0.046\ H^+ + 0.046\ e^- \rightarrow 0.0058\ CH_3COO^- + 0.0172\ H_2O$$

$$0.158R_{\text{prop}}:\ 0.0226\ CO_2 + 0.0113\ HCO_3^- + 0.158\ H^+ + 0.158\ e^- \rightarrow 0.0113\ CH_3CH_2COO^- + 0.0564\ H_2O$$

$$R_a:\ 0.128\ CO_2 + 0.017\ HCO_3^- + H^+ + e^- \rightarrow 0.0995\ CH_4 + 0.0058\ CH_3COO^- + 0.0113\ CH_3CH_2COO^- + 0.273\ H_2O$$

The donor reaction is constructed similarly:

$$0.33R_{\text{lact}}:\ 0.055\ CO_2 + 0.0275\ HCO_3^- + 0.33\ H^+ + 0.33\ e^- \rightarrow 0.0275\ CH_3CHOHCOO^- + 0.11\ H_2O$$

$$0.67R_{\text{glu}}:\ 0.168\ CO_2 + 0.67\ H^+ + 0.67\ e^- \rightarrow 0.0279\ C_6H_{12}O_6 + 0.168\ H_2O$$

$$R_d:\ 0.223\ CO_2 + 0.0275\ HCO_3^- + H^+ + e^- \rightarrow 0.0275\ CH_3CHOHCOO^- + 0.0279\ C_6H_{12}O_6 + 0.278\ H_2O$$

Finally, we use the relationship that $R_e = R_a - R_d$ to obtain the overall balanced energy reaction:

$$0.0275CH_3CHOHCOO^- + 0.0279C_6H_{12}O_6 + 0.005H_2O \rightarrow 0.0995CH_4 + 0.0058CH_3COO^- + 0.0113CH_3CH_2COO^- + 0.095CO_2 + 0.0105HCO_3^-$$

A quick mass-balance check shows that the different species are consistent with the problem statement if it is assumed that 1 mol each of lactate and glucose is converted to the reduced end products. However, 1.1 mol of each was consumed. The difference is due to net biomass synthesis. If this is true, then $f_s = 0.1/1.1$ or 0.091, and $f_e = 1 - 0.091$ or 0.909.

The overall reaction including energy and synthesis is constructed by adding 0.909 of the energy reaction we just formed to 0.091 of the synthesis reaction from the donors to biomass. The result is the following:

$$0.0275\ CH_3CHOHCOO^- + 0.0279\ C_6H_{12}O_6 + 0.0046\ NH_4^+ \rightarrow 0.0046\ C_5H_7O_2N + 0.0904\ CH_4 + 0.00527\ CH_3COO^- + 0.0103\ CH_3CH_2COO^- + 0.088\ CO_2 + 0.0075\ HCO_3^- + 0.011\ H_2O$$

Such an equation would be difficult to construct from the information provided without the formal procedure as presented here.

The approach used to analyze fermentations is applicable for analyzing any situation in which reduced products are formed. An excellent example is sulfate reduction, in which sulfate is reduced to sulfide, just as methane is in the reaction above. If reduced organic products are present along with the sulfide, then the above procedure can be used to construct an energy reaction or overall biological reaction in a similar manner. An example of such an energy reaction is lactate conversion to acetate by *Desulfovibrio desulfuricans* through sulfate reduction:

$$0.084CH_3CHOHCOO^- + 0.042SO_4^{2-} + 0.063H^+ \rightarrow$$
$$0.084CH_3COO^- + 0.021H_2S + 0.021HS^- + 0.084CO_2 + 0.084H_2O$$

The reaction shows that $e_{\text{acetate}} = 0.67$ and $e_{\text{sulfide}} = 0.33$ for this case.

2.6 Energetics and Bacterial Growth

Microorganisms carry out oxidation-reduction reactions in order to obtain energy for growth and cell maintenance. The amount of energy released per electron equivalent of an electron donor oxidized varies considerably from reaction to reaction. It is not surprising then that the amount of growth that results from an equivalent of donor oxidized varies considerably as well. The purpose of this section is to explore relationships between reaction energetics and bacterial growth.

Figure 1.13 is a simple illustration of how cells capture and transport energy that is released from a catabolic oxidation-reduction reaction to synthesize new cells or to use in cell maintenance. Cell maintenance has energy requirements for activities such as cell movement and repair of cellular proteins that decay because of normal resource recycling or through interactions with toxic compounds. When cells grow rapidly in the presence of nonlimiting concentrations of all factors required for growth, cells make the maximum investment of energy for synthesis. However, when an essential factor, such as the electron-donor substrate, is limited in concentration, then a larger portion of the energy obtained from substrate oxidation must be used for cell maintenance. This is illustrated by Equation 2.6, which indicates that the net yield of cells decreases with a decrease in the rate of substrate utilization. Equation 2.7 further indicates that the net yield becomes zero when the energy supplied through substrate utilization is just equal to m, the maintenance energy. Under these conditions, all energy released is used just to maintain the integrity of the cells. If the substrate available for the microorganisms decreases further, then food is insufficient to maintain the microorganisms, and they go into net decay. Conversely, when substrate and all other required factors are unlimited in amount, the rate of substrate utilization will be at its maximum, and the net yield, Y_n, will approach (but not quite reach) the true yield, Y.

Over the years, many approaches have been taken to describe how the true yield relates to the energy released from electron-donor oxidation. An extensive summary, historical review, and discussion of relationships presented between energy production and cell yield are provided by Battley (1987). Although no unified approach for relating growth to reaction energetics is yet widely accepted, a method that is based on electron equivalents and that differentiates between the energy portion of an overall biological reaction and the synthesis portion, as exemplified by Equation 2.37, has proven highly useful. This is the approach that is used here, and it is based on broader discussions presented elsewhere (McCarty, 1971, 1975; Christensen and McCarty, 1975).

In addition to being fundamentally sound, a practical advantage of the approach presented here is that electron equivalents are easily related to measurements of widespread utility in environmental engineering practice, that is, the common expression of waste strength in terms of oxygen demand, or OD. Examples are the biochemical oxygen demand (BOD), calculated oxygen demand (COD′), and chemical oxygen demand (COD). Since one equivalent of oxygen is 8 g of O_2, one equivalent of any electron donor is equivalent to an OD of 8 g as O_2. Thus, numbers of equivalents per liter can be directly converted into an OD concentration for that substance. This makes it very easy to compute Y values that contain the widely used units of COD, COD′, or BOD.

Example 2.9

COMPUTING COD′ A wastewater contains 12.6 g/l of ethanol. Estimate the e^- eq/l and the COD′ (g/l) for this wastewater.

From reaction O-5 in Table 2.3, there are 12 e^- eq/mol ethanol. Since 1 mol of ethanol weighs 46 g, the equivalent weight is 46/12 or 3.83 g/e^- eq. The ethanol concentration in the wastewater is thus 12.6/3.83 or 3.29 e^- eq/l. The COD′ thus becomes (8 g OD/e^- eq)(3.29 e^- eq/l) = 26.3 g/l.

2.6.1 FREE ENERGY OF THE ENERGY REACTION

Tables 2.2 and 2.3 summarize the standard free energies corrected to pH 7 ($\Delta G^{0'}$) for various inorganic and organic half-reactions, respectively. A student can readily determine standard free energies of other half-reactions by using values of free energy of formation for individual constituents, as listed in Appendix A. For illustration, the half-reaction for oxidation of 2-chlorobenzoate is obtained in the following manner.

Balanced half-reaction for 2-chlorobenzoate formation:

$$\frac{1}{28}\ HCO^-_{3(aq)} + \frac{3}{14}\ CO_{2(g)} + \frac{1}{28}\ Cl^-_{(aq)} + \frac{29}{28}\ H^+_{(aq,\,10^{-7})} + e^-$$
$$\rightarrow \frac{1}{28}\ C_6H_4ClCOO^- + \frac{13}{28}\ H_2O_{(1)}$$

The free energies of formation for each species from Appendix A are (in kJ/e^- eq)

$$\frac{1}{28}(-586.85),\ \frac{3}{14}(-394.36),\ \frac{1}{28}(-31.35),\ \frac{29}{28}(-39.87),\ 0,$$
$$\frac{1}{28}(-237.9),\ \frac{13}{28}(-237.18)$$

The half-reaction free energy is calculated as the sum of the product free energies minus the sum of the reactant free energies, or $\Delta G^{0'} = 29.26$ kJ/e$^-$ eq.

In order to obtain the free energy for a full energy reaction, ΔG_r, the free energy for the donor half-reaction is added to the free energy for the acceptor half-reaction. This simple procedure is parallel to the way the overall energy reaction was constructed from half-reactions in the previous examples. The free energy for the donor half-reaction is the negative of the value listed in the tables, since a donor half-reaction must be written as the reverse of that listed in the table. For example, the aerobic oxidation of ethanol is constructed from half-reactions having standard free energy values in the tables.

		$\Delta G^{\circ'}$ kJ/e$^-$ eq
Reaction I-14:	$\frac{1}{4}O_2 + H^+ + e^- = \frac{1}{2}H_2O$	−78.72
− Reaction O-5:	$\frac{1}{12}CH_3CH_2OH + \frac{1}{4}H_2O = \frac{1}{6}CO_2 + H^+ + e^-$	−31.18
Result:	$\frac{1}{12}CH_3CH_2OH + \frac{1}{4}O_2 = \frac{1}{6}CO_2 + \frac{1}{4}H_2O$	−109.90

The resulting $\Delta G_r^{0'}$ of −109.90 kJ/e$^-$ eq is specifically for the standard conditions indicated in the tables (i.e., 1 M ethanol aqueous concentration, 1 atm partial pressures of oxygen and carbon dioxide, and liquid water). The pH also is fixed at 7.0. Since H^+ does not appear on either side of the overall ethanol oxidation equation, the fixed pH is of no consequence in this case. However, pH can have a significant effect in other cases.

Figure 2.2 illustrates the relationships among various electron donors and acceptors and the resulting reaction free energy, assuming all constituents are at unit activity except pH = 7.0. This figure illustrates that, for the range of organic materials from methane to glucose, $\Delta G_r^{0'}$ for aerobic oxidation (oxygen as electron acceptor) or denitrification (nitrate as electron acceptor) varies relatively little, covering a range from about −96 to −120 kJ/e$^-$ eq. With inorganic electron acceptors, on the other hand, the range is very large, varying from about −5 kJ/e$^-$ eq with iron oxidation to −119 kJ/e$^-$ eq with hydrogen oxidation under aerobic conditions. However, if one considers organic electron donors under anaerobic conditions, either with carbon dioxide or sulfate as electron acceptors, the relative range of $\Delta G_r^{0'}$ is also quite large. For example, with carbon dioxide as the electron acceptor (methanogenesis), $\Delta G_r^{0'}$ is −3.87 kJ/e$^-$ eq for acetate oxidation and −17.82 kJ/e$^-$ eq for glucose oxidation, a factor of 4.6 difference. The great differences in reaction free energies for aerobic versus anaerobic and organic versus inorganic reactions have great effects on resulting bacterial yields, as will be demonstrated later.

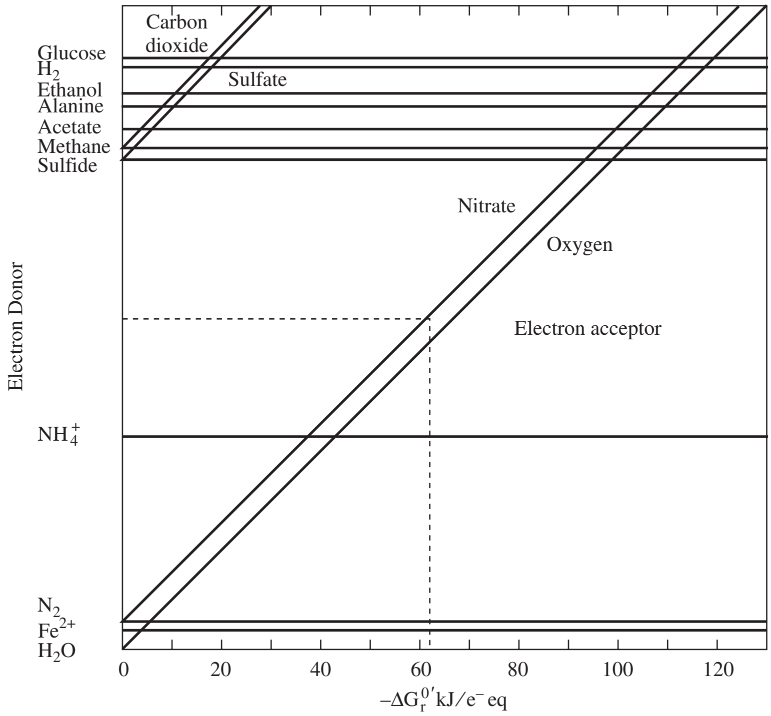

Figure 2.2 Relationship between various electron donors and acceptors and resulting reaction free energy.

One can adjust the reaction free energy for nonstandard concentrations of reactants and products. We first consider a generic reaction r that involves n different constituents:

$$v_1 A_1 + v_2 A_2 + \cdots \rightarrow v_m A_m + v_{m+1} A_{m+1} + \cdots + v_n A_n \qquad \textbf{[2.40]}$$

which can be written in an even more general form:

$$0 = \sum_{i=1}^{n} v_{ir} A_i \qquad \textbf{[2.41]}$$

The value of v_{ir} is negative if constituent A_i appears on the left side of Equation 2.40 and positive if it appears on the right side.

The nonstandard free energy change for this reaction can be determined from

$$\Delta G_r = \Delta G_r^0 + RT \sum_{i=1}^{n} v_{ir} \ln a_i \qquad \textbf{[2.42]}$$

Here, v_{ir} represents the stoichiometric coefficient for constituent A_i in reaction r, and a_i represents the activity of constituent A_i. T is absolute temperature (K).

The terms are easily identified from the final reaction of the ethanol example, which is repeated here,

$$\frac{1}{12}\ CH_3CH_2OH + \frac{1}{4}\ O_2 = \frac{1}{6}\ CO_2 + \frac{1}{4}\ H_2O$$

For this energy reaction, ΔG_r^0 equals $\Delta G_r^{0\prime}$ because H^+ is not a component of the equation; $n = 4$; and v_{ik} equals $-1/12$, $-1/4$, 1/6, and 1/4 for ethanol, oxygen, carbon dioxide, and water, respectively. Substituting the information into Equation 2.42 gives

$$\Delta G_r = \Delta G_r^0 + RT \ln \frac{[CO_2]^{1/6}[H_2O]^{1/4}}{[CH_3CH_2OH]^{1/12}[O_2]^{1/4}} \qquad \textbf{[2.43]}$$

We assume that the concentration of ethanol in an aqueous solution is 0.002 M, the oxygen partial pressure is that in the normal atmosphere at sea level (0.21 atm), the carbon dioxide concentration is that in normal air (0.0003 atm), and the temperature is 20 °C. We also assume that the activities of the three constituents are equal to their molar concentration or partial pressure. (These values could be corrected if one knew the respective activity coefficients for each constituent. In this case the activity coefficient is likely to be close to 1.0.) With aqueous solutions in which water is the major solvent, the activity of H_2O is sufficiently close to 1. Then,

$$\Delta G_r = -109{,}900 + 8.314(273 + 20) \ln \frac{[0.0003]^{1/6}[1]^{1/4}}{[0.002]^{1/12}[0.21]^{1/4}}$$
$$= -111{,}000 \text{ J/e}^- \text{ eq or } -111 \text{ kJ/e}^- \text{ eq}$$

A conclusion from this exercise is that the reaction free energy, corrected for concentrations that are within a typical range for biological systems of interest, is within 1 percent of the standard free energy of -109.9 kJ/e$^-$ eq. This is frequently the case, providing a correction has been made for pH.

While the exercise with aerobic oxidation of ethanol indicates that corrections for constituent concentrations may not be necessary, this is not always the case. Concentration corrections become important when the concentration of the electron donor or acceptor is very low or when the magnitude of ΔG_r^0 for a reaction is on the order of -10 kJ/e$^-$ eq or less. The latter situation is common for some anaerobic electron acceptors and inorganic electron donors.

The $\Delta G^{0\prime}$ values in Tables 2.2 and 2.3 already have the pH correction for pH = 7.0. If pH is significantly different from 7, then corrections for the effect of H^+ concentration may be necessary. When pH is significantly different than 7, or in cases where corrections to $\Delta G^{0\prime}$ are deemed necessary, one can convert to ΔG^0, which considers $\{H^+\} = 1$, with

$$\Delta G_r^0 = \Delta G_r^{0\prime} - RT\, v_{H^+} \ln[10^{-7}] \qquad \textbf{[2.44]}$$

2.7 YIELD COEFFICIENT AND REACTION ENERGETICS

The microbial yield from substrate utilization takes place in two steps. First, the energy reaction creates high-energy carriers, such as ATP. Second, the energy carriers are "spent" to drive cell synthesis or cell maintenance (recall Figure 1.13). As with all reactions, a certain amount of thermodynamic free energy is lost with each transfer. In this section, we describe how to compute the energy costs of cell synthesis and how to account for the energy lost in transfers. By combining all three aspects, we are able to estimate f_s^0 and the true yield (Y) based on thermodynamic principles. For determining the true yield and in accordance with Equation 2.5, the energy for maintenance is set to zero, so that all energy is used for cell synthesis.

First, we specify that the energy required to synthesize one equivalent of cells from a given carbon source is ΔG_s when the nitrogen source is ammonium. We first must determine the energy change resulting from the conversion of the carbon source to the common organic intermediates that the cell uses for synthesizing its macromolecules. We use pyruvate as a representative intermediate. The half-reaction for pyruvate is given as reaction O-16 in Table 2.3, and it has a free energy of 35.09 kJ/e$^-$ eq. The energy required to convert the carbon source to pyruvate is ΔG_p and is computed as the difference between the free energy of the pyruvate half-reaction and that of the carbon source:

$$\Delta G_p = 35.09 - \Delta G_c^{0'} \qquad \textbf{[2.45]}$$

For heterotrophic bacteria, the carbon source almost always is the electron donor. Thus, $\Delta G_c^{0'}$ is a value taken from Table 2.3 for the given electron donor. For example, if the electron donor were acetate, $\Delta G_c^{0'}$ would equal 27.4 kJ/e$^-$ eq. In autotrophic reactions, inorganic carbon is used as the carbon source. Considerable energy is required to reduce inorganic carbon to pyruvate. In photosynthesis, the hydrogen or electrons for reducing carbon dioxide to form cellular organic matter comes from water. By analogy to this case, we can determine the energy involved if we take $\Delta G_c^{0'}$ to equal that for the water-oxygen reaction from Table 2.2, or -78.72 kJ/e$^-$ eq. Thus, ΔG_p for the autotrophic case is always taken to equal 35.09 – (-78.72) = 113.8 kJ/e$^-$ eq.

Next, pyruvate carbon is converted to cellular carbon. The energy required here (ΔG_{pc}) is based on an estimated value of 3.33 kJ per gram cells (McCarty, 1971). From Table 2.4, an electron equivalent of cells is 113/20 = 5.65 grams when ammonia is used as the nitrogen source. Thus, ΔG_{pc} is $3.33 \cdot 5.65 = 18.8$ kJ/e^- eq for the case with ammonium nitrogen.

Finally, energy is always lost in the electron transfers. The loss is considered by including a term for energy-transfer efficiency ε. In sum, the energy requirement for cell synthesis becomes

$$\Delta G_s = \frac{\Delta G_p}{\varepsilon^n} + \frac{\Delta G_{pc}}{\varepsilon} \qquad \textbf{[2.46]}$$

We note that an exponent n is used for the energy transfer efficiency for conversion of carbon to pyruvate. The n accounts for the fact that ΔG_p for some electron donors, such as glucose, is negative, meaning energy is obtained by its conversion to pyruvate. Some of this energy is lost, and for such cases, n is taken to equal –1. In other cases, such as with acetate, ΔG_p is positive, meaning that energy is required in its conversion to pyruvate. Here, more energy is required than the thermodynamic amount, and for this case, $n = +1$.

Now that we have an estimate of how much energy is needed to synthesize an equivalent of cells, we can estimate how much electron donor must be oxidized to supply the energy. We define that, in order to supply this amount of energy, A equivalents of electron donor must be oxidized. The energy released by this oxidation is $A\Delta G_r$, where ΔG_r is the free energy released per equivalent of donor oxidized for energy generation. When this energy is transferred to the energy carrier, a portion again is lost through transfer inefficiencies. If the transfer efficiency here is the same as that for transferring energy from the carrier to synthesis (ε), then the energy transferred to the carrier will is $\varepsilon A \Delta G_r$.

An energy balance must be maintained around the energy carrier at steady state:

$$A\varepsilon\Delta G_r + \Delta G_s = 0 \tag{2.47}$$

Solving for A gives:

$$A = -\frac{\dfrac{\Delta G_p}{\varepsilon^n} + \dfrac{\Delta G_{pc}}{\varepsilon}}{\varepsilon \Delta G_r} \tag{2.48}$$

This equation indicates that the equivalents of donor used for energy production per equivalent of cells formed (A) increases as the energy required for synthesis from the given carbon source increases and as the energy released by donor oxidation decreases.

Our ultimate goal is developing balanced stoichiometric equations. As Equation 2.48 does not include energy of maintenance, the resulting value of A is for the situation of the true yield or Y. Thus, f_s is its maximum value or f_s^0, and the portion of donor used for energy is at its minimum value, f_e^0. Since part of the donor consumed is used for energy (A equivalents in this case) and the other part for synthesis (1.0 equivalents in this case), the total donor used is $1 + A$. Thus, we can compute f_s^0 and f_e^0 from A by

$$f_s^0 = \frac{1}{1+A} \quad \text{and} \quad f_e^0 = 1 - f_s^0 = \frac{A}{1+A} \tag{2.49}$$

The energy-transfer efficiency is the key factor that needs to be assumed in solving Equation 2.48. Under optimum conditions, transfer efficiencies of 55 to 70 percent are typical, and a ε value of 0.6 frequently is employed. However, some reported cell yields, especially with pure cultures that may be grown on substrates for which they do not have highly efficient enzyme systems, suggest transfer efficiencies lower than 0.6. In general, we would not expect microorganisms to survive well in mixed cultures if they had a low efficiency. On the other hand, some enzymatic reactions, such as the initial oxidation of hydrocarbons (information in Chapters 1 and 14), require an energy input, such as in the form of NADH, even though the overall reaction being carried out may result in net energy release. Those reactions should have a lower energy yield due to the initial energy investment, but these energetic calculations do not directly take into account such activation reactions.

Example 2.10

EFFECTS OF ε ON HETEROTROPHIC YIELD Compare estimates for f_s^0 and Y for aerobic oxidation of acetate, assuming $\varepsilon = 0.4$, 0.6, and 0.7, that pH = 7, and that all other reactants and products are at unit activity. Ammonium is available for synthesis.

Since the reaction is heterotrophic, Equation 2.45 applies. Using reaction O-1 from Table 2.3, $\Delta G_d^{0'} = 27.40$ kJ/e$^-$ eq. Therefore,

$$\Delta G_p = 35.09 - 27.40 = 7.69 \text{ kJ/e}^- \text{ eq}$$

Since this is an aerobic reaction, $\Delta G_a^{0'} = -78.72$ kJ/e$^-$ eq. Thus,

$$\Delta G_r = \Delta G_a^{0'} - \Delta G_d^{0'} = -78.72 - 27.40 = -106.12 \text{ kJ/e}^- \text{ eq}$$

Since ΔG_p is positive, $n = +1$. Also, since ammonium is available for cell synthesis, ΔG_{pc} equals 18.8 kJ/e^- eq. Hence,

$$A = -\frac{\dfrac{7.69}{\varepsilon^{+1}} + \dfrac{18.8}{\varepsilon}}{-106.12\varepsilon}$$

Letting $\varepsilon = 0.4$, 0.6, and 0.7, $A = 1.56$, 0.69, and 0.51, respectively. Using Equation 2.49, values for f_s^0 that result are:

ε	f_s^0	Y g cells/mol donor	Y g cells/g donor	Y g cells/g COD$'$
0.4	0.39	18	0.30	0.28
0.6	0.59	27	0.45	0.42
0.7	0.66	30	0.51	0.47

In order to determine bacterial yield, it is well to write a balanced stoichiometric equation. This will be done for illustration only for the case where $\varepsilon = 0.6$, for which $f_s^0 = 0.59$ and $f_e^0 = 1 - 0.59 = 0.41$:

$-$ Reaction O-1: $0.125\ CH_3COO^- + 0.375\ H_2O \rightarrow 0.125\ CO_2 + 0.125\ HCO_3^- + H^+ + e^-$

0.41(Reaction I-14): $0.1025\ O_2 + 0.41\ H^+ + 0.41\ e^- \rightarrow 0.205\ H_2O$

0.59(Reaction C-1): $0.118\ CO_2 + 0.0295\ HCO_3^- + 0.0295\ NH_4^+ + 0.59\ H^+ + 0.59\ e^- \rightarrow 0.0295\ C_5H_7O_2N + 0.2655\ H_2O$

Overall: $0.125\ CH_3COO^- + 0.1025\ O_2 + 0.0295\ NH_4^+ \rightarrow 0.0295\ C_5H_7O_2N + 0.007\ CO_2 + 0.0955\ HCO_3^- + 0.0955\ H_2O$

From this balanced reaction:

$$\begin{aligned} Y &= 0.0295(113)/0.125 = 27 \text{ g cells/mol acetate} \\ &= 0.0295(113)/0.125(59) = 0.45 \text{ g cells/g acetate} \\ &= 0.0295(113)/8 = 0.42 \text{ g cells/g COD}' \end{aligned}$$

Example 2.11

EFFECTS OF ELECTRON ACCEPTORS AND DONORS FOR HETEROTROPHS Compare estimated values of f_s^0 for glucose and acetate when oxygen, nitrate, sulfate, and carbon dioxide are the electron acceptors. The calculated values will equal the total net synthesis, which may include more than one microbial type. Assume $\varepsilon = 0.6$ and that the nitrogen source is ammonium.

The $\Delta G_d^{0'}$ and $\Delta G_a^{0'}$ values are in Tables 2.3 and 2.4, respectively. Since ammonium is the nitrogen source, ΔG_{pc} is 18.8 kJ/e$^-$eq.

Acetate ($\Delta G_d^{0'}$ = 27.40 kJ/e$^-$ eq; ΔG_p = 7.69 kJ/e$^-$ eq; n = +1):

Electron Acceptor	$\Delta G_a^{0'}$	ΔG_r	A	f_s^0
Oxygen	−78.72	−106.12	0.69	0.59
Nitrate	−72.20	−99.60	0.74	0.57
Sulfate	+20.85	−6.55	11.2	0.08
CO_2	+23.53	−3.87	19.0	0.05

Glucose ($\Delta G_d^{0'}$ = 41.35 kJ/e$^-$ eq; ΔG_p = −6.26 kJ/e$^-$ eq; n = -1):

Electron Acceptor	$\Delta G_a^{0'}$	ΔG_r	A	f_s^0
Oxygen	−78.72	−120.07	0.38	0.72
Nitrate	−72.20	−113.55	0.40	0.71
Sulfate	+20.85	−20.50	2.24	0.31
CO_2	+23.53	−17.82	2.58	0.28

The yield, as indicated by f_s^0, is lower for acetate than for glucose. Carbohydrates have more positive $\Delta G_d^{0'}$ values due to their more ordered (lower entropy) structure. This makes ΔG_p negative. Oxygen and nitrate give much higher yields than do sulfate and carbon dioxide, which is due to the different values of $\Delta G_a^{0'}$. These trends are consistent with and help explain empirical findings for the different types of microorganisms.

Example 2.12

YIELDS FOR VARIOUS AEROBIC CHEMOLITHOTROPHS Compare estimated f_s^0 values for bacterial chemolithotrophs that oxidize ammonium to nitrate, sulfide to sulfate, Fe(II) to Fe(III), and H_2 to H_2O under aerobic conditions. Carbon dioxide and ammonium are the C and N sources, respectively. Assume $\varepsilon = 0.6$.

Since all the microorganisms are autotrophs (carbon dioxide as C source), ΔG_s is the same: $113.8/0.6 + 18.8/0.6 = 221$ kJ/e$^-$ eq. The values of $\Delta G_d^{0'}$ come from Table 2.2, while $\Delta G_a^{\circ'} = -78.72$ kJ/e$^-$ eq for O_2. The results are:

Electron Donor	$\Delta G_d^{0'}$	ΔG_r	A	f_s^0
Ammonium	−35.11	−43.61	8.45	0.11
Sulfide	+20.85	−99.57	3.70	0.21
Fe(II)	−74.27	−4.45	82.8	0.012
Hydrogen	+39.87	−118.59	3.11	0.24

Compared to the f_s^0 values in Example 2.11, all values for the chemolithotrophs are low. This comes about mainly from the high cost of autotrophic biomass synthesis, which is shown quantitatively by the large ΔG_s. Despite the low yields, all of these electron donors are exploited by chemolithotrophs under aerobic conditions. Interestingly, the low yield with iron oxidation is increased under low-pH conditions when a correction is made to ΔG_r for pH (Equation 2.44). Iron oxidizers that are tolerant to very low pH (e.g., 2) are well known.

2.8 Oxidized Nitrogen Sources

Microorganisms prefer to use ammonium nitrogen as an inorganic nitrogen source for cell synthesis, because it already is in the (−III) oxidation state, the status of organic nitrogen within the cell. However, when ammonium is not available for synthesis, many prokaryotic cells can use oxidized forms of nitrogen as alternatives. Included here are nitrate (NO_3^-), nitrite (NO_2^-), and dinitrogen (N_2). When an oxidized form of nitrogen is used, the microorganisms must reduce it to the (−III) oxidation state of ammonium, a process that requires electrons and energy, thus reducing their availability for synthesis.

Using an oxidized nitrogen source affects the energy cost of synthesis, or ΔG_s. We extend the energetics method presented in the previous section to include the added synthesis costs. The accounting approach assumes that all electrons used for synthesis are first routed to the common organic intermediate, pyruvate. Thus, the ΔG_p part of ΔG_s remains the same (Equation 2.45):

$$\Delta G_p = 35.09 - \Delta G_c^{0'}$$

The electrons needed to reduce the oxidized form of nitrogen to ammonium are assumed to come from pyruvate and transferred to the oxidized nitrogen source. Reducing nitrogen for synthesis is not part of respiration, and it does not yield any energy. Therefore, reducing the nitrogen source is an energy cost in that the electrons that otherwise could have been transferred to the acceptor to give energy are diverted to nitrogen reduction for synthesis. The ΔG_{pc} part of ΔG_s depends on the nitrogen source, since a mole of $C_5H_7O_2N$ contains different numbers of electron equivalents, depending on how many electrons had to be invested in nitrogen reduction: 20 e$^-$ eq/mole for NH_4^+, 28 for NO_3^-, 26 for NO_3^-, and 23 for N_2. The number of electron equivalents invested in a mole of cells can be seen from half-reactions C-1 to C-4 in Table 2.4. The energy cost to synthesize cells from the common intermediates (i.e., ΔG_{pc}) is the same in units of kJ/gram, that is, 3.33 kJ/g cells (McCarty, 1971), but it varies with the number of electron equivalents per mole of cells: $\Delta G_{pc} = 18.8$ kJ/e$^-$ eq for NH_4^+, 13.5 kJ/e$^-$ eq for NO_3^-, 14.5 kJ/e$^-$ eq for NO_2^-, and 16.4 kJ/e$^-$ eq for N_2. Equation 2.48 is used to compute A as before, but the value of ΔG_{pc} in the numerator depends on the nitrogen source, as shown in the previous sentence.

Once A is computed from Equation 2.48, f_s^0 and f_e^0 are computed in the usual manner from Equation 2.49. Then, the overall stoichiometric reaction is generated in the normal manner from the appropriate donor (R_d), acceptor (R_a), and synthesis (R_c) half-reactions. The key is that the synthesis half-reaction be selected from C-2 to C-4, instead of C-1 (Table 2.4), when the nitrogen source is oxidized. The entire process is illustrated for nitrate as the nitrogen source in the following example, which shows that using an oxidized nitrogen source decreases the true yield (Y) due to the electron and energy investments to reduce it.

Example 2.13 **EFFECT OF AN OXIDIZED NITROGEN SOURCE** Estimate f_s^0 and Y (in g cells/g COD′) for acetate utilization under aerobic conditions when NO_3^- is used as the source of nitrogen for cell synthesis. Write the stoichiometric equation for the overall reaction. Determine the oxygen requirement in g O_2/g COD′, and compare with the result from Example 2.10, when NH_4^+ was used for cell synthesis. Assume $\varepsilon = 0.6$.

From Example 2.10, $\Delta G_p = 7.69$ kJ/e$^-$ eq, $n = +1$, and $\Delta G_r = -106.12$ kJ/e$^-$ eq. When nitrate is the nitrogen source, $\Delta G_{pc} = 13.5$ kJ/e$^-$ eq. Then,

$$A = -\frac{\dfrac{7.69}{0.6^{+1}} + \dfrac{13.5}{0.6}}{0.6(-106.12)} = 0.55$$

and

$$f_s^0 = \frac{1}{1 + A} = \frac{1}{1 + 0.55} = 0.65$$

With $f_e^0 = 1 - f_s^0 = 0.35$, the overall reaction for biological growth is then developed as follows:

$$
\begin{array}{ll}
0.35R_a: & 0.0875\ O_2 + 0.35\ H^+ + 0.35\ e^- \rightarrow 0.175\ H_2O \\
0.65R_c: & 0.0232\ NO_3^- + 0.1161\ CO_2 + 0.6732\ H^+ + 0.65\ e^- \rightarrow \\
 & \qquad 0.0232\ C_5H_7O_2N + 0.2554\ H_2O \\
-R_d: & 0.125\ CH_3COO^- + 0.375\ H_2O \rightarrow 0.125\ CO_2 + 0.125\ HCO_3^- + H^+ + e^- \\
\hline
\text{Net:} & 0.125\ CH_3COO^- + 0.0875\ O_2 + 0.0232\ NO_3^- + 0.0232\ H^+ \rightarrow \\
 & \qquad 0.0232\ C_5H_7O_2N + 0.0089\ CO_2 + 0.125\ HCO_3^- + 0.0554\ H_2O
\end{array}
$$

From this equation,

$$\begin{aligned} Y &= 0.0232(113)/0.125 = 21 \text{ g cells/mol acetate} \\ &= 0.0232(113)/0.125(59) = 0.36 \text{ g cells/g acetate} \\ &= 0.0232(113)/8 = 0.33 \text{ g cells/g COD}' \end{aligned}$$

The oxygen requirement is 0.0875 (32)/8 = 0.35 g O_2/g COD′.

For comparison, when ammonium is the nitrogen source:

$$Y = 0.42 \text{ g cells/g COD}'$$

$$\text{Oxygen required} = 0.1025(32)/8 = 0.41 \text{ g } O_2/\text{g COD}'$$

The comparison indicates that, when nitrate is used as the nitrogen source, the yield of bacterial cells and the requirement for oxygen are less than when ammonia is the nitrogen source. The diversion of electrons from acetate for the reduction of nitrate to ammonium for cell synthesis reduces the acetate available for energy generation and synthesis.

2.9 Bibliography

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2.10 Problems

2.1. Which of the following electron donor/electron acceptor pairs represent potential energy reactions for bacterial growth? Assume that all reactants and products are at unit activity, except that pH = 7.

Case	Electron Donor	Electron Acceptor
a.	Acetate	carbon dioxide (methanogenesis)
b.	Acetate	Fe^{3+} (Reduction to Fe^{2+})
c.	Acetate	H^+ (reduction to H_2)
d.	Glucose	H^+ (reduction to H_2)
e.	H_2	carbon dioxide (methanogenesis)
f.	H_2	nitrate (denitrification to N_2)
g.	S (oxidized to sulfate)	NO_3^- (denitrification to N_2)
h.	CH_4	NO_3^- (denitrification to N_2)
i.	NH_4^+ (oxidation to NO_2^-)	SO_4^{2-} (reduction to $H_2S + HS^-$)

2.2. Estimate f_s^o for each of the following, assuming all constituents are at unit activity, except that pH = 7.0, and that the energy transfer efficiency (ε) is 0.6:

Case	Electron Donor	Electron Acceptor	Nitrogen Source
a.	Ethanol	oxygen	ammonia
b.	Ethanol	oxygen	nitrate
c.	Ethanol	sulfate	ammonia
d.	Ethanol	carbon dioxide (methanogenesis)	ammonia
e.	Propionate	carbon dioxide (methanogenesis)	ammonia
f.	Sulfur (oxidized to sulfate)	nitrate (denitrification to nitrogen gas)	ammonia
g.	Ammonia (oxidized to nitrite)	oxygen	ammonia

2.3. Organic matter is converted in sequential steps by different bacterial species to methane in anaerobic methanogenesis of organic wastes. One important step is the conversion of the intermediate butyrate to acetate, for which the following electron donor and acceptor half-reactions apply:

Electron donor:

$$\frac{1}{2}\underset{\text{acetate}}{CH_3COO^-} + \frac{1}{4}CO_2 + H^+ + e^- = \frac{1}{4}\underset{\text{butyrate}}{CH_3CH_2CH_2COO^-} + \frac{1}{4}HCO_3^- + \frac{1}{4}H_2O$$

Electron acceptor half-reaction:

$$H^+ + e^- = \frac{1}{2} H_{2(g)}$$

Determine ΔG_r for the resulting energy reaction under the following conditions:

1. (*a*) All constituents are at unit activity
   (*b*) All constituents are at unit activity, except pH = 7.0
   (*c*) The following typical activities under anaerobic conditions apply:

$$[CH_3COO^-] = 10^{-3}M, [CO_2] = 0.3 \text{ atm}, pH = 7.0,$$
$$[CH_3CH_2CH_2COO^-] = 10^{-2}M, \text{ and } [HCO_3^-] = 10^{-1}M.$$
$$P_{H2(g)} = 10^{-6} \text{ atm}.$$

2. Under which of the above three conditions is it possible for bacteria to obtain energy for growth?

2.4. Per equivalent of electron donor oxidized, would you expect to obtain more cell production from anaerobic conversion of ethanol to methane or from oxidation of acetate through reduction of sulfate to sulfide? Why?

2.5. Estimate through use of reaction energetics the yield coefficient Y in g cells per gram substrate for benzoate when used as an electron donor and when sulfate is the electron acceptor (sulfate reduced to sulfide). Assume ammonia is present for cell synthesis and that the energy transfer efficiency (ε) equals 0.6.

2.6. How many grams of oxygen are required per electron equivalent of lactate in a wastewater when subjected to aerobic biological treatment if f_s is 0.4 (assume sufficient ammonium is present for cell synthesis)?

2.7. For each of the following, write an oxidation half-reaction and normalize the reaction on an electron-equivalent basis. Add H_2O as appropriate to either side of the equations in balancing the reactions.

(*a*) $CH_3CH_2CH_2CHNH_2COO^-$ oxidation to CO_2, NH_4^+, and HCO_3^-
(*b*) Cl^- to ClO_3^-

2.8. You wish to consider the addition of methanol to groundwater for anaerobic biological removal of nitrate by denitrification. If the nitrate nitrogen concentration is 84 mg/l, what minimum concentration of methanol should be added to achieve complete nitrate reduction to nitrogen gas? Assume no ammonium is present and that f_s for the reaction is 0.30.

2.9. Some facultative bacteria can grow on acetate while using tetrachloroethylene (PCE) as an electron acceptor, and that here, $f_s = 0.16$. Determine the energy-transfer efficiency for cell synthesis considering the following to be the electron acceptor reaction by which PCE is reduced to cis-1,2-dichloroethylene:

$$\frac{1}{4} CCl_2{=}CCl_2 + \frac{1}{2} H^+ + e^- = \frac{1}{4} CHCl{=}CHCl + \frac{1}{2} Cl^-$$

$$\Delta G^{\circ'} = -53.55 \text{ kJ/e}^- \text{ eq}$$

2.10. Use energetics to compute f_s° and Y (in g cells/g COD′) for the four situations in which the electron donor is citrate, the electron acceptor is oxygen, the carbon source is citrate, and the nitrogen source is either ammonium, nitrate, nitrite, or dinitrogen. Assume $\varepsilon = 0.6$.

2.11. Repeat problem 2.10, except that methane replaces citrate as the electron donor and carbon source.

2.12. Nitrification is carried out by autotrophs that utilize ammonium as the electron donor, oxygen as the electron acceptor, ammonium as the nitrogen source, and inorganic carbon as the carbon source. First, you are to use energetics to compute f_s° and Y (g cells per g NH_4^+-N) for this normal case when ammonium is oxidized to nitrate. Second, compute the changes to f_s° and Y if the bacteria were able to use acetate as the carbon source, instead of inorganic carbon. Assume $\varepsilon = 0.6$.

2.13. Use energetics to compute f_s° and Y (g cells per g H_2) for hydrogen-oxidizing bacteria that use either oxygen or sulfate as the electron acceptor (two sets of answers). In both cases, the cells are autotrophs. Assume $\varepsilon = 0.6$.

2.14. Write the balanced, overall reaction for the situation in which acetate is the donor and carbon source, nitrate is the acceptor and nitrogen source, and $f_s = 0.333$.

2.15. Write the balanced, overall reaction for the situation in which Cu(s) is the donor (it is oxidized to Cu^{2+}), oxygen is the acceptor, inorganic carbon is the carbon source, ammonium is the nitrogen source, and $f_s^\circ = 0.036$.

2.16. Write the balanced overall reaction for the situation in which ammonium is the donor and is oxidized to nitrite, oxygen is the acceptor, the cells are autotrophs, ammonium in the nitrogen source, and $f_e^\circ = 0.939$.

2.17. Write the balanced overall reaction for the situation in which dichloromethane (CH_2Cl_2) is the donor and carbon source, oxygen is the acceptor, ammonium is the nitrogen source, and $f_s^\circ = 0.31$.